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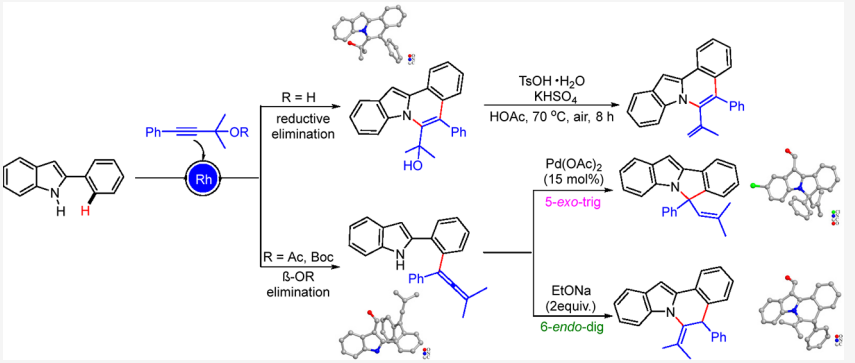
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Diversity-Oriented Synthesis of Indole-Fused Polycyclic Scaffolds via Rhodium-Catalyzed *NH*-Indole-Directed C−H Coupling of 2-Phenyl-1*H*-indoles with Propargylic Alcohol Derivatives

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ABSTRACT: Diversity-oriented synthesis strategy for the efficient assembly of indole-fused polycyclic scaffolds via rhodium-catalyzed *NH*-indole-directed C−H coupling with propargylic alcohol derivatives in a regioselective manner was developed. Five 2-phenyl-1*H*-indole-embedded core skeletons were synthesized. In particular, three different indole-fused exo-olefin-containing polycycles were realized, which may be manipulated for further chemistry.

INTRODUCTION

Indole-fused polycyclic scaffolds have been found in many natural or unnatural compounds, showing a broad range of structural diversities and biological activities.[1] In recent years, diversity-oriented synthesis (DOS) has been an efficient method for building libraries of small molecules with both functional groups and skeletal diversity.[2] Consequently, the development of new diversity-oriented synthesis for the collections of indole-fused polycyclic molecules has gained significant interest among synthetic chemists.[3] On the other hand, metal-catalyzed C−H activations have emerged as a powerful tool to synthesize functionalized complex molecules in a sustainable manner.[4] Recently, our group[5] and others[6] have successfully developed the strategy of *NH*-indole-directed C−H activation/annulation for the assembly of indole-fused polycyclic structures via the reaction of 2-phenyl-1*H*-indoles or 7-phenyl-1*H*-indoles with various coupling partners. Unfortunately, the coupling partners reported are limited to simple monofunctional reagents, such as carbon monoxide,[5b] alkenes,[5aceh6g] alkynes,[5g6abf] or diazo compounds,[5df6h] which inevitably lead to the poor structural diversity of target products.

Propargylic alcohol derivatives are easily accessible from terminal alkynes and ketones or aldehydes.[7] The presence of the alkyne moiety and the OR unit in propargylic alcohol derivatives endows diversified reactivity and an ideal synthetic precursor for diversity-oriented synthesis.[8] As shown in Scheme 1, metal-catalyzed directing group-assisted C−H coupling with propargylic alcohol derivatives includes the following common process. (1) Metal species **A** cleaves the *ortho*-C−H bond of the substrate to generate metallacycle **B**.[9] (2) Migratory insertion of propargylic alcohol derivative into **B** affords another metallacycle **C**. The weak coordination of an oxygen atom with a metal catalyst would greatly benefit for excellent region and chemoselectivity.[10] (3) **C** would undergo the reductive elimination to release the annulated product **D**.[11] (4) *β*-Hydroxyl elimination of intermediate C would also happen to produce allene **F**,[10c12] which can further undergo intramolecular annulation to obtain various cyclic compounds due to allene's multiple reactive sites.[13] In this context, propargylic alcohol derivatives have attracted considerable attention recently for building a variety of cyclic molecules.[14] Bearing these in mind and as a continuation of our interest in

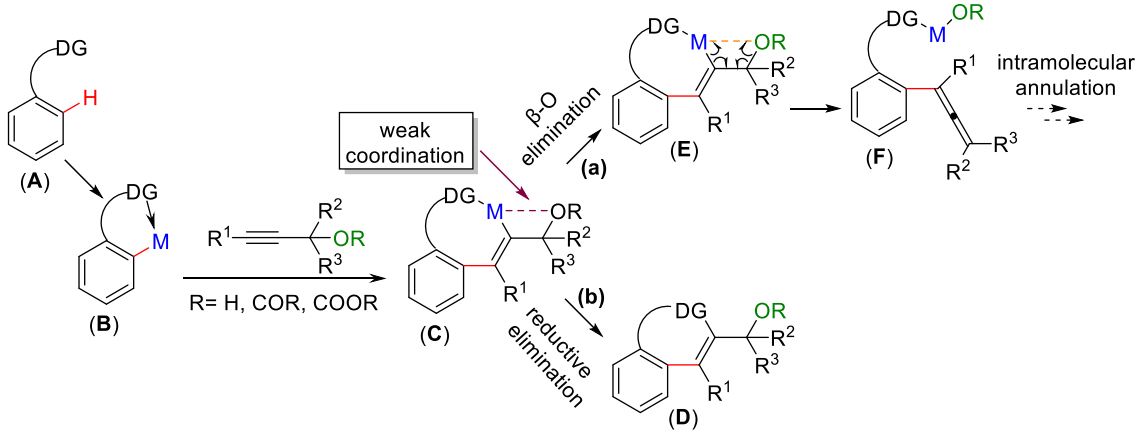
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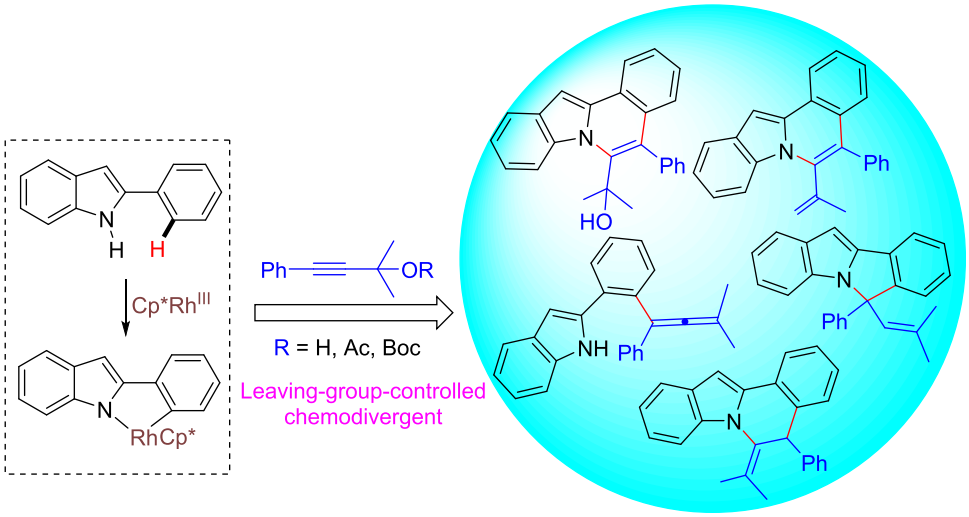
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Scheme 1. Metal-Catalyzed Directing Group-Assisted C−H Coupling with Propargylic Alcohol Derivatives



Scheme 2. Rh-Catalyzed C−H Coupling of 2-Phenyl-1*H*-Indoles with Propargylic Alcohol Derivatives



metal-catalyzed C−H functionalization of indoles, herein, we disclose rhodium-catalyzed *NH*-indole-directed C−H coupling of 2-phenyl-1*H*-indoles with propargylic alcohol derivatives, furnishing 2-(2-(3-methyl-1-phenylbuta-1,2-dien-1-yl)phenyl)-1*H*-indoles, 5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo-[2,1-*a*]isoquinolines, 6-(2-methylprop-1-en-1-yl)-6-phenyl-6*H*-isoindolo[2,1-*a*]indoles, 2-(5-phenylindolo[2,1-*a*]isoquinolin-6-yl)propan-2-ols, and 5-phenyl-6-(prop-1-en-2-yl)indolo[2,1-*a*]isoquinolines (Scheme 2).

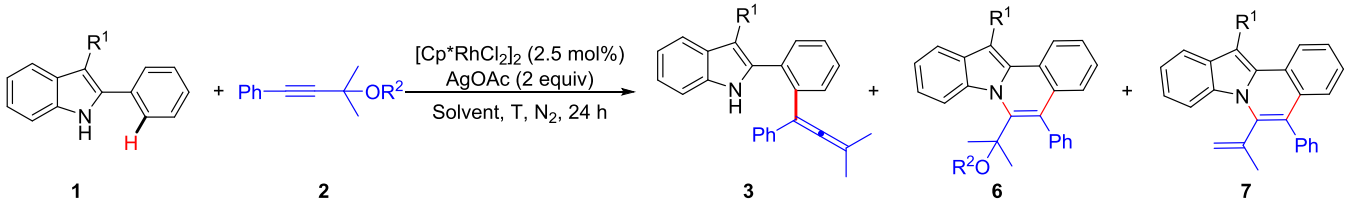
RESULTS AND DISCUSSION

Initially, we attempted the coupling reaction of the model substrates 2-phenyl-1*H*-indole-3-carbaldehyde **1a** and *tert*-butyl (2-methyl-4-phenylbut-3-yn-2-yl) carbonate **2a**. Encouragingly, we found that $[Cp^*RhCl_2]_2$ was capable of catalyzing the reaction to produce the mixture of *ortho*-allenylated-2-phenyl-1*H*-indole **3a** and 5-phenyl-6-(prop-1-en-2-yl)indolo-[2,1-*a*]isoquinoline-12-carbaldehyde **7a** in yields of 31 and 8%, respectively, using toluene as the solvent (Table 1, entry 1). Further screening of other solvents indicated that EtOAc is the most efficient solvent for selective allenylation to afford **3a** in 71% yield (entries 2−6). The studies on the influence of temperature showed that high temperature favored the formation of **7a** (entries 7−11). Performing the reaction at 100 °C resulted in a 42% yield of **7a** (entry 10). However, further increasing the temperature to 120 °C led to a messy reaction mixture and provided only a trace of **3a** or **7a** (entry 11). Similar results were observed when the reaction was carried out between **1a** and 2-methyl-4-phenylbut-3-yn-2-yl acetate (**2b**). The product **3a** was isolated with 68% yield (entry 12). For 2-methyl-4-phenylbut-3-yn-2-ol (**2c**), the allenylated product **3** was not observed at all. Instead, the [4 + 2] oxidative annulation product **6** was obtained in 20% yield exclusively (entry 14). Many attempts to improve the yield of **6** were unsuccessful. Interestingly, the yield of **6** significantly increased to 80% when the formyl group was removed from substrate **1** (entry 15). Finally, subsequent acid-mediated dehydration of **6** provided access to **7** in excellent yield (83%). Control experiments showed that *N*-methyl substituted 2-phenyl-1*H*-indole was unreactive in the process (entry 16).

The generality of 2-phenyl-1*H*-indoles for the allenylation was first investigated, and the results are shown in Scheme 3. The allenylation can be scaled up to the 5 mmol scale with the optimal reaction conditions, and **3a** was isolated in 56% yield (1.1 g). The 2-phenyl-1*H*-indoles bearing groups at the meta- and para-positions of the benzene ring underwent the allenylation smoothly to give the corresponding products **3b**−**3i** in good yields. In contrast, *ortho*-substituted substrate **1** afforded products **3j**−**3l** in relatively low yields due to the increased steric hindrance. For 2-phenyl-1*H*-indole, no

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Table 1. Optimization of the Reaction Conditions[a]



entry	substrate **1**	substrate **2**	solvent	temperature	yield (%)[b] **3**	**6**	**7**
1	R^1=CHO	**2a**, R^2=Boc	toluene	60 °C	**3a**, 31	0	**7a**, 8
2	R^1=CHO	**2a**, R^2=Boc	MeOH	60 °C	**3a**, 26	0	**7a**, 7
3	R^1=CHO	**2a**, R^2=Boc	CH_3CN	60 °C	**3a**, 23	0	**7a**, 9
4	R^1=CHO	**2a**, R^2=Boc	DMF	60 °C	**3a**, 0	0	**7a**, 0
5	R^1=CHO	**2a**, R^2=Boc	DMSO	60 °C	**3a**, 0	0	**7a**, 0
6	R^1=CHO	**2a**, R^2=Boc	EtOAc	60 °C	**3a**, **71**	0	**7a**, 14
7	R^1=CHO	**2a**, R^2=Boc	EtOAc	RT	**3a**, 29	0	**7a**, 0
8	R^1=CHO	**2a**, R^2=Boc	EtOAc	45 °C	**3a**, 47	0	**7a**, trace
9	R^1=CHO	**2a**, R^2=Boc	EtOAc	80 °C	**3a**, 24	0	**7a**, 34
10	R^1=CHO	**2a**, R^2=Boc	EtOAc	100 °C	**3a**,trace	0	**7a**, 42
11	R^1=CHO	**2a**, R^2=Boc	EtOAc	120 °C	**3a**,trace	0	**7a**, trace
12	R^1=CHO	**2b**, R^2=Ac	EtOAc	60 °C	**3a**, 68	0	**7a**, 5
13	R^1=CHO	**2c**, R^2=H	EtOAc	60 °C	0	trace	0
14	R^1=CHO	**2c**, R^2=H	EtOAc	100 °C	0	**6a**, 20	0
15[c]	R^1=H	**2c**, R^2=H	EtOAc	100 °C	0	**6b**, **80**	0
16[d]	R^1=H	**2a** or **2c**	EtOAc	60 or 100 °C	0	0	0

[a]Reaction conditions: **1** (0.2 mmol), **2** (0.3 mmol, 1.5 equiv), $[Cp^*RhCl_2]_2$ (2.5 mol %) AgOAc (0.4 mmol, 2 equiv), solvent (2.0 mL), N_2, T, 24 h. [b]Isolated yield. [c]**1** (0.5 mmol), **2c** (0.5 mmol), $[Cp^*RhCl_2]_2$ (2 mol %), AgOAc (1 mmol), Na_2CO_3 (1 mmol), EtOAc (3 mL), 6 h, isolated yield. [d]1-Methyl-2-phenyl-1*H*-indole was used as the starting material.

reaction occurred, and the starting material was recovered. When 2,3-diphenyl-1*H*-indole or 3-methyl-2-phenyl-1*H*-indole was employed as the substrate, the reaction would give rise to an inseparable complicated mixture. Notably, when substrates with electron-withdrawing or electron-donating groups at the indole ring were employed, the reactions worked well to give products **3m**−**3r** in good yields. The reaction of 2-phenyl-1*H*-indole-3-carbaldehyde **1a** and *tert*-butyl (2-methyl-4-phenylbut-3-yn-2-yl) carbonate **2d** afforded the desired product **3r** in 61% yield. As can be seen from Scheme 3, the *NH*-indole-directed C−H allenylation exhibited good tolerance to F, Cl, Br, CHO, OMe, CO_2Me, and NO_2 groups. Compound **3a** could be easily further reduced or oxidized, giving **3aa** or **3ab** without disrupting the allenyl motif.

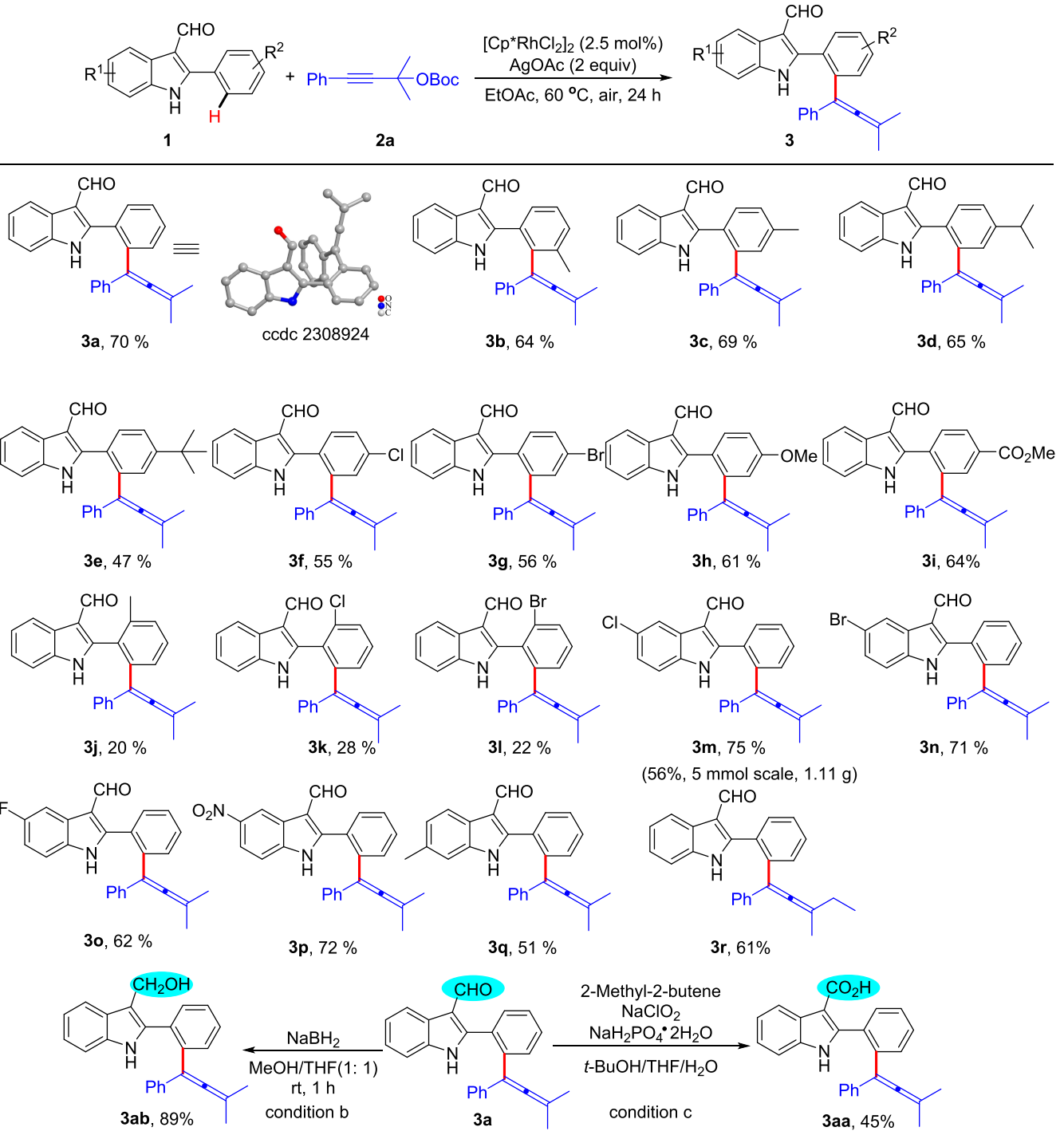
Next, the efficient construction of versatile indole-fused *exo*-olefin-containing polycyclic compounds was realized by taking advantage of the inherent multiple reactivities of the allenyl motif (Scheme 4).[5ach6g] When *ortho*-allenylated-2-phenyl-1*H*-indole **3a** was treated with EtONa in *N*,*N*-dimethylformamide (DMF) at 60 °C for 5 h, we observed exclusively 6-*endo*-dig cyclization to give 5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo[2,1-*a*]isoquinoline **4a** in excellent yield, whose structure was confirmed by X-ray crystal diffraction. The exploration of substrate scope, as shown in the left column of Scheme 4, revealed that the base-mediated 6-*endo*-dig hydroamination exhibited brilliant compatibility with various substituents of **3**. The yield of the corresponding product was up to 97% (**4a**−**4o**). The 6-*endo*-dig cyclization of **3r** delivered a mixture of products **4p** + **4p′** in total 82% yield. When the cyclization of **3a** was performed in the presence of the $Pd(OAc)_2$ catalyst in cyclohexane at 100 °C, a 5-*exo*-trig annulation product **5a** instead of a 6-*endo*-dig product was isolated. The structure of the 5-*exo*-trig annulation product was unambiguously established by single-crystal X-ray analysis of compound **5i**. *Ortho*-allenylated-2-phenyl-1*H*-indoles featuring different substituents proceeded to give the desired products **5a**−**5f** and **5k**−**5l** in moderate to good yields. Whereas the yields were reduced to roughly 40% when the Cl or Br group was present in the starting material **3** (**5g**−**5j**). Not surprisingly, the 5-*exo*-trig annulation of **3r** afforded a mixture of products **5m** + **5m′** (34% yield).

At the end, the oxidative annulation/dehydration cascade reaction of 2-phenyl-1*H*-indoles with 2-methyl-4-phenylbut-3-yn-2-ol was examined (Scheme 5). In most cases, 2-(5-phenylindolo[2,1-*a*]isoquinolin-6-yl)propan-2-ols **6a**−**6m** and subsequent 5-phenyl-6-(prop-1-en-2-yl)indolo[2,1-*a*]-isoquinolines products, **7a**−**7m**, were obtained in moderate to very good yields and in a highly regioselective manner. The oxidative annulation can be scaled up to the 4 mmol scale with the optimal reaction conditions, and **6j** was isolated in 51% yield (835 mg). It is worth noting that the Cl, Br, CN, NO_2, or CO_2Me group remained intact after the oxidative annulation/dehydration cascade process, showing an excellent functional group tolerance.

Based on the above results and previous literature studies,[11,12] a plausible mechanism of the coupling of 2-phenyl-1*H*-indoles with propargylic alcohol derivatives is proposed (Scheme 6). Initially, N−H bond cleavage and subsequent C−H bond cleavage by the Cp*Rh(III) catalyst gives five-membered rhoda(III)cycle **II**. Then, the migratory insertion of propargylic alcohol derivatives into the Rh−C bond in a regioselective manner via the weak coordination between the rhodium atom and the oxygen atom provides a seven-membered rhoda(III)cycle **III**. Intermediate **III** under-

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Scheme 3. Substrate Scope for the Synthesis of 3[a]



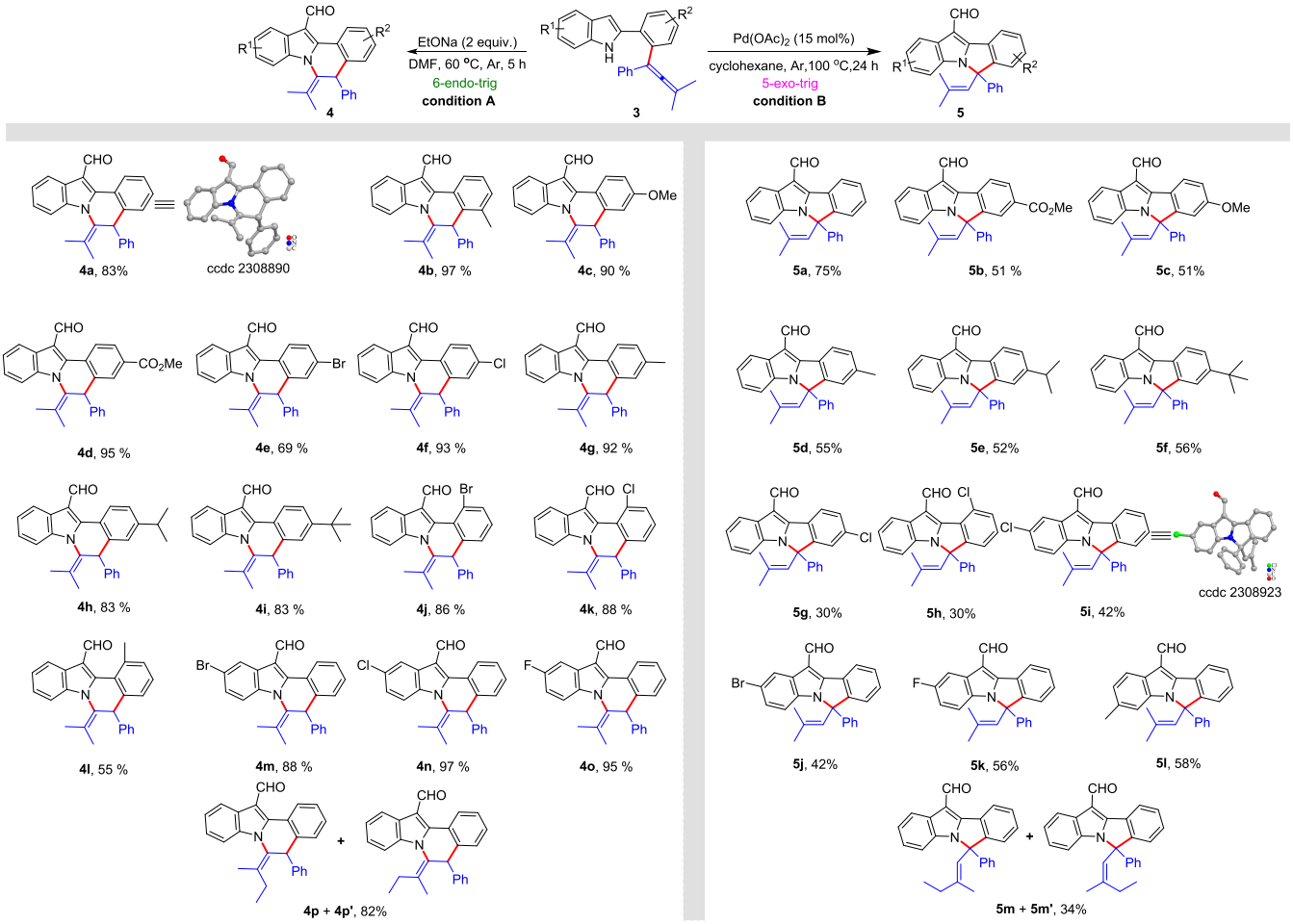
[a]Reaction conditions: **1** (0.2 mmol), **2a** (0.3 mmol), $[Cp^*RhCl_2]_2$ (2.5 mol %), AgOAc (0.4 mmol), EtOAc (2.0 mL), air, 60 °C, 24 h; isolated yield; Condition b: **3a** (0.2 mmol), $NaBH_4$ (0.4 mmol), MeOH/THF = 1:1 (1.0 mL), rt, 1 h; isolated; Condition c: **3a** (0.3 mmol), 2-Methyl-2-butene (550 μL), $NaClO_2$ (1.5 mol), $NaH_2PO_4{\cdot}2H_2O$ (4.5 mmol), *t*-BuOH/THF/H_2O = 1:1:1 (1.8 mL), air, rt, 24 h; isolated.

goes reductive elimination to give Cp*Rh(I) species and releases product **6**, which can further dehydrate to afford product **7** (route 2). Meanwhile, intermediate **III** could also proceed via β-OR elimination to form intermediate **IV**, followed by protonation and reductive elimination to liberate allenyl-substituted product **3** (route 1). Finally, the active catalyst Cp*Rh(III) regenerates in the presence of a Ag(I) oxidant.

The mechanisms of two independent allenyl transformations are also presented in Scheme 7. N-Deprotonation of the indole with EtONa leads to the formation of the activated nucleophile indole-N^-, which attacks the C2 of the allene motif to give 6-*endo*-dig product **4**. On the other hand, in the case of 5-*exo*-trig cyclization, coordination of the active Pd(0) catalyst with the indole N atom gives the Pd(II)-H intermediate **Int-3-5-I**,[15] followed by the insertion of the allene into the Pd−H bond to

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Scheme 4. Substrate Scope for the Synthesis of 4 and 5



[a]Reaction conditions: Condition A, **3** (0.2 mmol), EtONa (0.4 mmol, 2 equiv), DMF (2.0 mL), Ar, 60 °C, 5 h, isolated yield; Condition B, 3 (0.2 mmol), $Pd(OAc)_2$ (15 mol %), cyclohexane (2.0 mL), Ar, 100 °C, 24 h; isolated yield.

generate the π-allyl-Pd(II) intermediate **Int-3-5-II**, which undergoes reductive elimination to release product **5**.

In conclusion, we have successfully developed rhodium-catalyzed *NH*-indole-directed C−H coupling with propargylic alcohol derivatives for the diversity-oriented synthesis of indole-fused polycyclic scaffolds. The reaction exhibits switchable regioselectivity depending on the propargylic alcohol derivatives. With propargylic alcohol (free OH) as the coupling partner, the reaction involves migratory insertion/reductive elimination to generate 2-(5-phenylindolo[2,1-*a*]isoquinolin-6-yl)propan-2-ols, which further undergo acid-mediated dehydration to deliver 5-phenyl-6-(prop-1-en-2-yl)indolo[2,1-*a*]-isoquinolines. Migratory insertion/β-OR elimination is involved in the coupling reaction to afford *ortho*-allenylated-2-phenyl-1*H*-indoles when the propargylic ester is used as the coupling partner. Thereafter, taking advantage of the inherent multiple reactivities of the allenyl motif, *ortho*-allenylated-2-phenyl-1*H*-indoles can selectively convert into 5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo[2,1-*a*]isoquinolines via base-mediated 6-*endo*-dig cyclization. On the other hand, 6-(2-methylprop-1-en-1-yl)-6-phenyl-6*H*-isoindolo[2,1-*a*]indoles could be obtained by palladium-catalyzed 5-*exo*-trig cyclization.

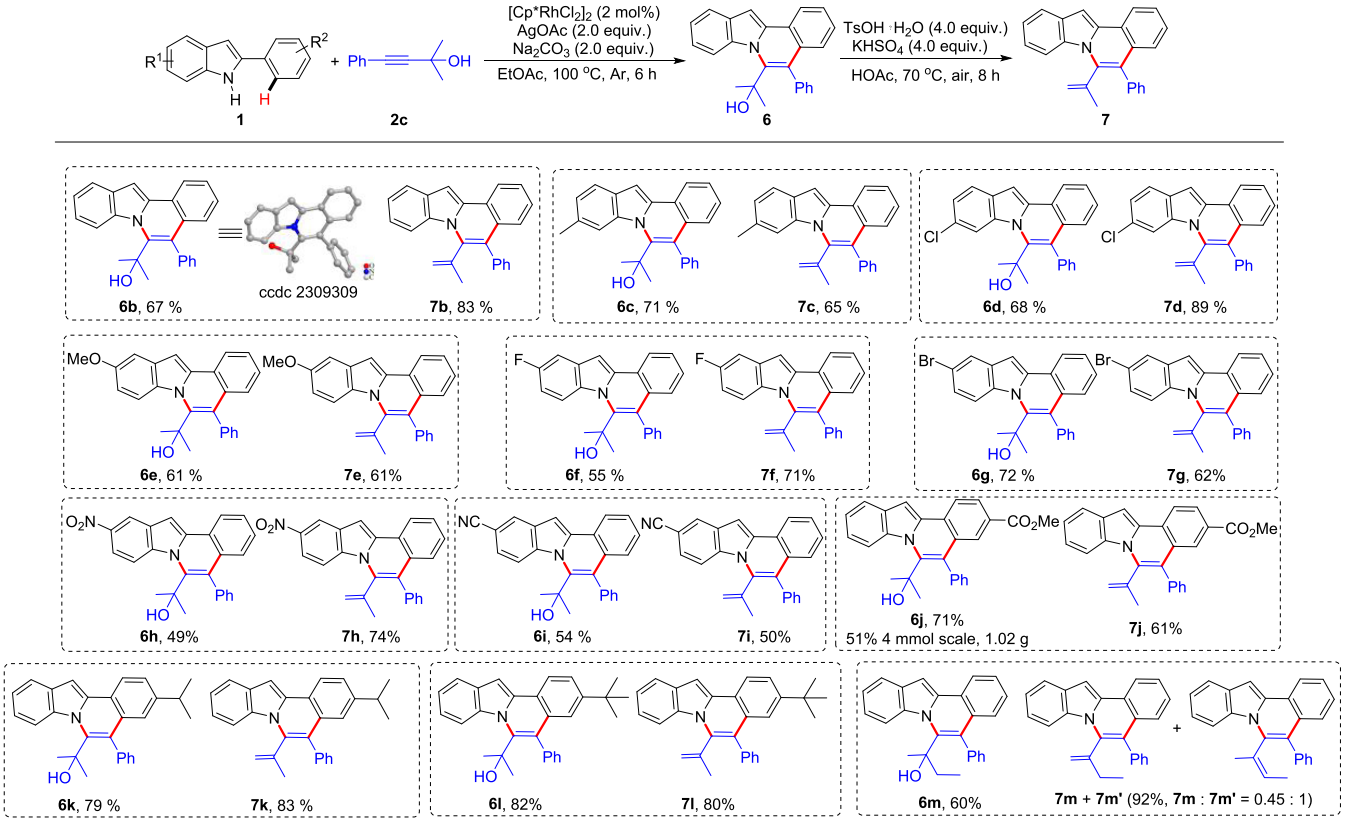
EXPERIMENTAL SECTION

General Method. Unless otherwise noted, the reagents (chemicals) were purchased from commercial sources and were used without further purification. 2-Phenyl-1*H*-indole (**1q**), 1*H*-indole, and phenylboronic acid are commercially available. 2-Aryl-1*H*-indoles (**1r**, **1s**, **1v**, **1w**, **1y**, **1z**, **1aa**) were prepared according to the literature procedures.[5a] 2-Aryl-1*H*-indoles (**1t**, **1u**, **1x**) were synthesized according to literature procedures.[16] **1r**−**1aa** are known compounds, and the ^{1}H NMR data of the isolated products were in agreement with the literature reports. 2-Phenyl-1*H*-indole-3-carbaldehydes (**1a**−**1p**) were synthesized from the literature procedures.[5d] **1a**−**1p** are known compounds, and ^{1}H NMR data of the isolated products were in agreement with the literature reports. Propargylic alcohol (**2c**) and propargylic alcohol derivatives (**2a, 2b, 2d, 2e**) were synthesized from the literature procedures.[17] **2a**−**2e** are known compounds, and ^{1}H NMR data of the isolated products were in agreement with the literature reports. ^{1}H NMR spectra were recorded at 400 MHz (600 MHz) and ^{13}C NMR spectra at 100 MHz (150 MHz), respectively. ^{19}F NMR spectra were recorded at 565 MHz. ^{1}H chemical shifts (δ) were referenced to TMS, and ^{13}C NMR chemical shifts (δ) were referenced to internal solvent resonance. Electrospray ionization high-resolution mass spectrometry (ESI-HRMS) spectra were recorded by using a Q-TOF mass spectrometer. Melting points (mp) were determined in open capillaries without further correction.

*2-Phenyl-1H-indole-3-carbaldehyde (**1a**).* Pink solid, 1.9 g (86%).[5d] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.43 (s, 1H), 9.96 (s, 1H), 8.21 (d, J = 7.2 Hz, 1H), 7.85−7.73 (m, 3H), 7.64−7.57 (m, 4H), 7.53−7.48 (m, 2H), 7.33−7.21 (m, 3H).

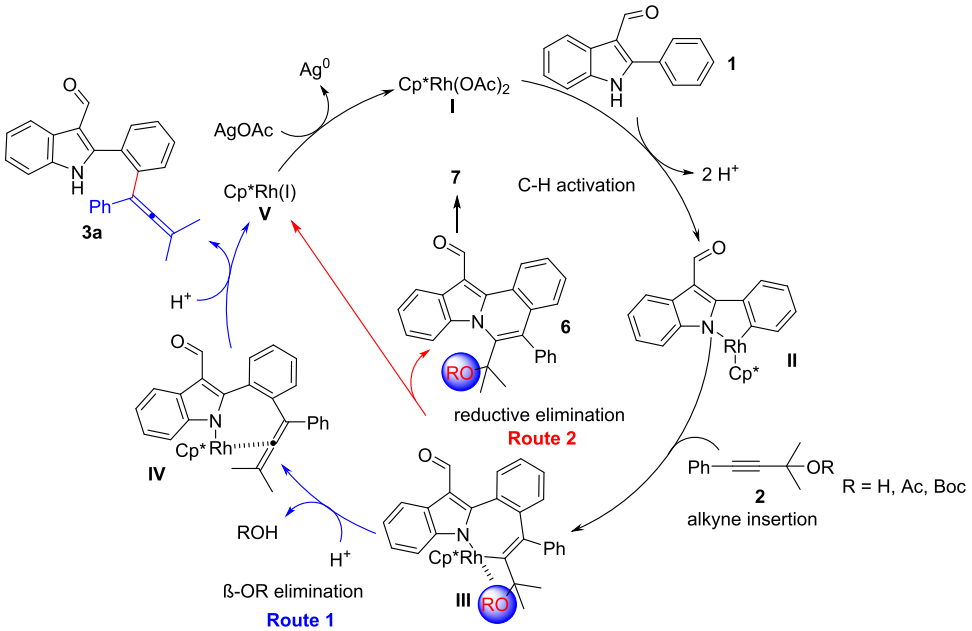
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Scheme 5. Substrate Scope for the Synthesis of 6 and 7[a]



[a]Reaction conditions: **1** (0.5 mmol), **2c** (0.5 mmol), $[Cp^*RhCl_2]_2$ (2 mol %), AgOAc (1 mmol), Na_2CO_3 (1 mmol), EtOAc (3.0 mL), 100 °C, Ar, 6 h, isolated yield; **6** (0.2 mmol), TsOH·H_2O (0.8 mmol), $KHSO_4$ (0.8 mmol), HOAc (1.0 mL), 70 °C, air, 8 h; isolated yield.

Scheme 6. Plausible Mechanism of the Coupling of 2-Phenyl-1*H*-indoles with Propargylic Alcohol Derivatives



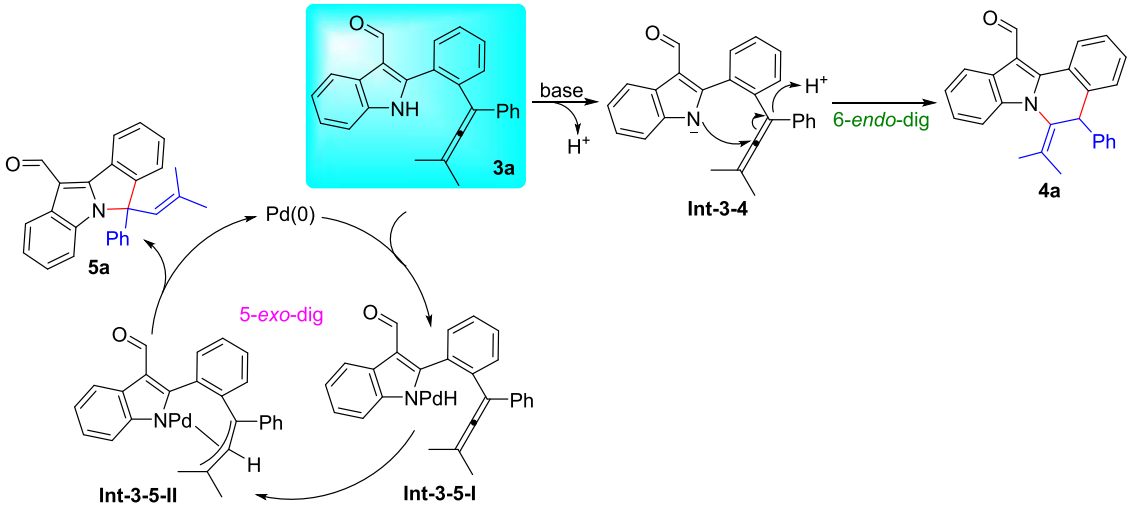
*2-(M-Tolyl)-1H-indole-3-carbaldehyde (**1b**).* Pink solid, 524 mg (86%).[5d] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.37 (s, 1H), 9.97 (s, 1H), 8.21 (d, J = 7.2 Hz, 1H), 7.61−7.55 (m, 2H), 7.52−7.46 (m, 2H), 7.39 (d, J = 7.6 Hz, 1H), 7.31−7.22(m, 2H), 2.43 (s, 3H).

*2-(P-Tolyl)-1H-indole-3-carbaldehyde (**1c**).* Pink solid, 1.1 g (85%).[5d] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.33 (s, 1H), 9.96 (s, 1H), 8.20 (d, J = 7.2 Hz, 1H), 7.67 (d, J = 8.0 Hz, 2H), 7.50 (d, J = 7.6 Hz, 1H), 7.41 (d, J = 8.0 Hz, 2H), 7.28 (td, J = 7.2, 1.2 Hz, 1H), 7.23 (td, J = 8.0, 1.6 Hz, 1H), 2.41 (s, 3H).

*2-(4-Isopropylphenyl)-1H-indole-3-carbaldehyde (**1d**).* Pink solid, 752 mg (79%).[5d] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.35 (s, 1H), 9.97 (s, 1H), 8.21 (d, J = 7.6 Hz, 1H), 7.70 (d, J = 8.0 Hz, 2H), 7.48 (t, J = 10.4 Hz, 3H), 7.26 (dt, J = 14.0, 7.2 Hz, 2H), 2.99 (m, 1H), 1.26 (d, J = 7.2 Hz, 6H).

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Scheme 7. Plausible Mechanisms of Allenyl Transformations



*2-(4-(tert-Butyl)phenyl)-1H-indole-3-carbaldehyde (**1e**).* Pink solid, 999 mg (88%).[18] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.39 (s, 1H), 9.98 (s, 1H), 8.21 (d, J = 6.8 Hz, 1H), 7.72 (d, J = 7.6 Hz, 2H), 7.62 (d, J = 7.6 Hz, 2H), 7.50 (d, J = 7.6 Hz, 1H), 7.26 (dq, J = 16.0, 6.8 Hz, 2H), 1.34 (s, 9H).

*2-(4-Chlorophenyl)-1H-indole-3-carbaldehyde (**1f**).* Yellow solid, 1.2 g (97%).[18] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.47 (s, 1H), 9.96 (s, 1H), 8.22 (d, J = 7.6 Hz, 1H), 7.81 (d, J = 8.4 Hz, 2H), 7.67 (d, J = 8.4 Hz, 2H), 7.52 (d, J = 7.6 Hz, 1H), 7.32−7.23 (m, 2H).

*2-(4-Bromophenyl)-1H-indole-3-carbaldehyde (**1g**).* Pink solid, 233 mg (77%).[18] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.44 (s, 1H), 9.96 (s, 1H), 8.22 (d, J = 7.2 Hz, 1H), 7.82−7.79 (m, 2H), 7.76−7.72 (m, 2H), 7.52 (d, J = 8.0 Hz, 1H), 7.30 (td, J = 7.2,1.2 Hz, 1H), 7.25 (td, J = 8.4, 1.2 Hz, 1H).

*2-(4-Methoxyphenyl)-1H-indole-3-carbaldehyde (**1h**).* Purple solid, 487 mg (97%).[5d] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.28 (s, 1H), 9.95 (s, 1H), 8.20 (d, J = 7.2 Hz, 1H), 7.75−7.71 (m, 2H), 7.48 (d, J = 7.6 Hz, 1H), 7.29−7.20 (m, 2H), 7.18−7.14 (m, 2H), 3.85 (s, 3H).

*Methyl-4-(3-formyl-1H-indol-2-yl)benzoate (**1i**).* Pink solid, 692 mg (52%).[5d] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.58 (s, 1H), 10.01 (s, 1H), 8.24 (d, J = 7.6 Hz, 1H), 8.15 (d, J = 8.4 Hz, 2H), 7.94 (d, J = 8.4 Hz, 2H), 7.54 (d, J = 8.0 Hz, 1H), 7.35−7.24 (m, 2H), 3.91 (s, 3H).

*2-(O-Tolyl)-1H-indole-3-carbaldehyde (**1j**).* Pink solid, 643 mg (57%).[5d] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.34 (s, 1H), 9.61 (s, 1H), 8.19 (d, J = 7.2 Hz, 1H), 7.51−7.42 (m, 4H), 7.37 (t, J = 7.3 Hz, 1H), 7.27 (p, J = 7.2, 6.7 Hz, 2H), 2.28 (s, 3H).

*2-(2-Chlorophenyl)-1H-indole-3-carbaldehyde (**1k**).* Pink solid, 1.0 g (77%).[5d] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.44 (s, 1H), 9.66 (s, 1H), 8.19 (d, J = 7.2 Hz, 1H), 7.74−7.70 (m, 2H), 7.62 (td, J = 7.6, 1.6 Hz, 1H), 7.57−7.50 (m, 2H), 7.33−7.25(m, 2H).

*5-Chloro-2-phenyl-1H-indole-3-carbaldehyde (**1l**).* Pink solid, 362 mg (94%).[5d] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.61 (s, 1H), 9.95 (s, 1H), 8.19 (d, J = 2.4 Hz, 1H), 7.81−7.78 (m, 2H), 7.64−7.58 (m, 3H), 7.53 (d, J = 8.8 Hz, 1H), 7.31 (dd, J = 8.8, 2.4 Hz, 1H).

*5-Bromo-2-phenyl-1H-indole-3-carbaldehyde (**1m**).* Pink solid, 260 mg (87%).[18] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.61 (s, 1H), 9.95 (s, 1H), 8.34 (s, 1H), 7.81−7.78 (m, 2H), 7.63−7.59 (m, 3H), 7.48 (d, J = 8.4 Hz, 1H), 7.44−7.41 (m, 1H).

*5-Fluoro-2-phenyl-1H-indole-3-carbaldehyde (**1n**).* Pink solid, 757 mg (70%).[18] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.54 (s, 1H), 9.96 (s, 1H), 7.89 (dd, J = 9.6, 2.0 Hz, 1H), 7.82−7.76 (m, 2H), 7.63−7.58 (m, 3H), 7.54−7.50 (m, 1H), 7.18−7.13 (m, 1H).

*5-Nitro-2-phenyl-1H-indole-3-carbaldehyde (**1o**).* Yellow solid, 906 mg (81%).[5d] ^{1}H NMR (400 MHz, DMSO-d_6) δ 9.98 (s, 1H), 9.01 (d, J = 2.0 Hz, 1H), 8.10 (dd, J = 8.8, 2.4 Hz, 1H), 7.83 (dd, J = 7.5, 2.4 Hz, 2H), 7.64−7.58 (m, 4H).

*6-Methyl-2-phenyl-1H-indole-3-carbaldehyde (**1p**).* Pink solid, 310 mg (78%).[18] ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.27 (s, 1H), 9.94 (d, J = 2.4 Hz, 1H), 8.10−8.03 (m, 1H), 7.78−7.75 (m, 2H), 7.62−7.54 (m, 3H), 7.41−7.28 (m, 1H), 7.13−7.07 (m, 1H), 2.43 (s, 3H).

*6-Methyl-2-phenyl-1H-indole (**1q**).* White solid, 1.2 g (58%). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.20 (s, 1H), 7.67−7.63 (m, 2H), 7.52 (d, J = 8.0 Hz, 1H), 7.44 (t, J = 7.2 Hz, 2H), 7.31 (t, J = 7.6 Hz, 1H), 7.19 (s, 1H), 6.97 (d, J = 8.8 Hz, 1H), 6.83−6.72 (m, 1H), 2.48 (s, 3H).

*6-Chloro-2-phenyl-1H-indole (**1r**).* White solid, 636.8 mg (56%).[5a] ^{1}H NMR (400 MHz, DMSO-d_6) δ 11.70 (d, J = 23.6 Hz, 1H), 7.84 (t, J = 6.8 Hz, 2H), 7.55−7.40 (m, 4H), 7.33 (t, J = 7.6 Hz, 1H), 7.13−7.07 (m, 1H), 6.94−6.83 (m, 1H).

*5-Methoxy-2-phenyl-1H-indole (**1s**).* White solid, 534 mg (24%).[5a] ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.24 (s, 1H), 7.65 (dd, J = 8.8, 1.6 Hz, 2H), 7.48−7.40 (m, 2H), 7.37−7.18 (m, 2H), 7.10 (d, J = 2.4 Hz, 1H), 6.87 (dd, J = 8.8, 2.8 Hz, 1H), 6.77 (dd, J = 2.0, 0.8 Hz, 1H), 3.88 (s, 3H).

*5-Fluoro-2-phenyl-1H-indole (**1t**).* White solid, 367 mg (66%).[5a] ^{1}H NMR (400 MHz, DMSO-d_6) δ 11.66 (s, 1H), 7.86 (d, J = 7.6 Hz, 2H), 7.46 (t, J = 7.6 Hz, 3H), 7.40 (dd, J = 8.4, 4.4 Hz, 2H), 7.35−7.25 (m, 3H), 6.95 (td, J = 9.6, 2.4 Hz, 2H), 6.88 (s, 1H).

*5-Bromo-2-phenyl-1H-indole (**1u**).* White solid, 979 mg (43%).[5a] ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.37 (s, 1H), 7.75 (s, 1H), 7.65 (dd, J = 8.4, 1.2 Hz, 2H), 7.46 (t, J = 7.6 Hz, 2H), 7.38−7.33 (m, 1H), 7.27 (d, J = 1.2 Hz, 2H), 6.76 (d, J = 2.0 Hz, 1H).

*5-Nitro-2-phenyl-1H-indole (**1v**).* Yellow solid, 3.7 g (78%).[5a] ^{1}H NMR (400 MHz, DMSO-d_6) δ 9.99 (s, 1H), 8.99 (d, J = 2.4 Hz, 1H), 8.12 (dd, J = 9.2, 2.4 Hz, 1H), 7.82 (dd, J = 7.6, 4.0 Hz, 2H), 7.64−7.60 (m, 4H).

*2-Phenyl-1H-indole-5-carbonitrile (**1w**).* White solid, 138 mg (31%).[5a] ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.79 (s, 1H), 7.97 (s, 1H), 7.68 (d, J = 7.2 Hz, 2H), 7.54−7.45 (m, 3H), 7.44−7.36 (m, 2H), 6.88 (d, J = 1.2 Hz, 1H).

*Methyl-4-(1H-indol-2-yl)benzoate (**1x**).* White solid, 347 mg (15%).[5a] ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.48 (s, 1H), 8.10 (d, J = 8.4 Hz, 2H), 7.73 (d, J = 8.8 Hz, 2H), 7.65 (d, J = 8.0 Hz, 1H), 7.42 (d, J = 8.4 Hz, 1H), 7.23 (d, J = 7.2 Hz, 1H), 7.14 (t, J = 7.6 Hz, 1H), 7.00−6.92 (m, 2H), 3.95 (s, 4H).

*2-(4-Isopropylphenyl)-1H-indole (**1z**).* White solid, 752 mg (79%).[5a] ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.31 (s, 1H), 7.62 (dd, J = 11.2, 8.0 Hz, 3H), 7.40 (d, J = 8.0 Hz, 1H), 7.32 (d, J = 8.0 Hz, 2H), 7.24−7.16 (m, 1H), 7.15−7.10 (m, 1H), 6.80 (s, 1H), 3.01−2.91(m, 1H), 1.31 (s, 3H), 1.29 (s, 3H).

*2-(4-(tert-Butyl)phenyl)-1H-indole (**1aa**).* White solid, 347 mg (15%).[5a] ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.31 (s, 1H), 7.66 (d, J =

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7.6 Hz, 1H), 7.61 (d, J = 8.0 Hz, 2H), 7.49 (d, J = 8.4 Hz, 2H), 7.41 (d, J = 8.0 Hz, 1H), 7.22 (t, J = 7.2 Hz, 1H), 7.15 (t, J = 7.6 Hz, 1H), 6.83 (s, 1H), 1.39 (s, 9H).

*tert-Butyl (2-Methyl-4-phenylbut-3-yn-2-yl) Carbonate (**2a**).* White solid, 2.84 g (64%).[17] ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.44−7.40 (m, 2H), 7.31−7.27 (m, 3H), 1.78 (s, 6H), 1.51 (s, 9H).

*2-Methyl-4-phenylbut-3-yn-2-yl Acetate (**2b**).* Yellow oil, 1.15 g (91%).[17] ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.43 (dd, J = 6.7, 3.0 Hz, 2H), 7.29 (q, J = 3.8 Hz, 3H), 2.05 (s, 3H), 1.75 (s, 6H).

*2-Methyl-4-phenylbut-3-yn-2-ol (**2c**).* Yellow oil, 2.7 g (67%).[17] ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.44−7.39 (m, 2H), 7.32−7.28 (m, 3H), 2.21 (s, 1H), 1.62 (s, 6H).

*tert-Butyl (3-Methyl-1-phenylpent-1-yn-3-yl) Carbonate (**2d**).* White oil, 2.5 g (80%).[17] ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.43 (dd, J = 6.4, 2.8 Hz, 2H), 7.32−7.27 (m, 3H), 2.10 (m, 1H), 1.96 (m, 1H), 1.76 (s, 3H), 1.50 (s, 9H), 1.10 (t, J = 7.2 Hz, 3H).

*3-Methyl-1-phenylpent-1-yn-3-ol (**2e**).* Yellow oil, 2.2 g (90%).[17] ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.45−7.40 (m, 2H), 7.29 (d, J = 4.4 Hz, 3H), 2.38 (s, 1H), 1.80 (m, 2H), 1.58 (s, 3H), 1.11 (t, J = 7.2 Hz, 3H).

General Procedure for the Synthesis of 3a−3r. A mixture of 2-pheyl-1*H*-indole-3-carbaldehydes **1** (0.2 mmol), **2a** (0.3 mmol, 78.1 mg), $[Cp^*RhCl_2]_2$ (2.5 mol %, 3.2 mg), AgOAc (0.4 mmol, 2 equiv, 66.8 mg), and EtOAc (2 mL) in a 50 mL sealed tube was heated in an oil bath at 60 °C for 24 h under an air atmosphere. The reaction mixture was cooled to room temperature and concentrated under reduced pressure, and then the residue was subjected to silica gel column chromatography [eluting with petroleum ether (PE)/ethyl acetate (EA)] to afford the corresponding product (**3a−3r**).

*2-(2-(3-Methyl-1-phenyl-buta-1,2-dien-1-yl)phenyl)-1H-indole-3-carbaldehyde (**3a**).* White solid, mp = 181.8−182.3 °C, 50.9 mg (70%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.16 (s, 1H), 9.59 (s, 1H), 8.02 (d, J = 7.2 Hz, 1H), 7.63−7.58 (m, 2H), 7.56−7.52 (m, 1H), 7.43 (dd, J = 8.0, 1.2 Hz, 1H), 7.37−7.35 (m, 1H), 7.20 (dd, J = 7.2, 1.6 Hz, 1H), 7.18−7.13 (m, 3H), 7.10−7.04 (m, 3H), 1.45 (s, 6H). ^{13}C{^{1}H} NMR (100 MHz, DMSO-d_6) δ 203.4, 184.9, 149.6, 138.2, 137.5, 136.2, 132.0, 130.8, 130.1, 128.4, 127.6, 127.0, 126.8, 125.2, 123.4, 122.3, 120.8, 114.6, 111.9, 105.4, 99.0, 19.8. HRMS (ESI) m/z calcd for $C_{26}H_{22}NO$ $[M + H]^+$: 364.1696, found: 364.1703. It was crystallized from PE/EA.

*2-(3-Methyl-2-(3-methyl-1-phenylbuta-1,2-dien-1-yl)phenyl)-1H-indole-3-carbaldehyde (**3b**).* Yellow solid, mp = 194.7−195.3 °C, 48.3 mg (64%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 9.84 (s, 1H), 8.55 (s, 1H), 8.30 (dd, J = 4.4, 2.8 Hz, 1H), 7.39−7.33 (m, 3H), 7.25−7.17 (m, 5H), 7.16−7.08 (m, 3H), 2.44 (s, 3H), 1.47 (s, 6H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 203.4, 186.9, 149.8, 138.3, 137.3, 135.6, 135.4, 133.1, 131.3, 130.8, 129.5, 128.4, 127.0, 126.8, 125.5, 123.9, 122.8, 122.0, 115.9, 111.0, 104.9, 99.4, 21.2, 20.0. HRMS (ESI) m/z calcd for $C_{27}H_{24}NO$ $[M + H]^+$: 378.1852, found: 378.1858.

*2-(4-Methyl-2-(3-methyl-1-phenylbuta-1,2-dien-1-yl)phenyl)-1H-indole-3-carbaldehyde (**3c**).* Yellow solid, mp = 209.6−210.2 °C, 52.1 mg (69%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.04 (s, 1H), 9.58 (s, 1H), 8.01 (d, J = 7.2 Hz, 1H), 7.47 (d, J = 8.0 Hz, 1H), 7.34 (d, J = 6.8 Hz, 2H), 7.22 (s, 1H), 7.18−7.10 (m, 4H), 7.06 (t, J = 8.0 Hz, 3H), 2.39 (s, 3H), 1.43 (s, 6H).^{13}C{^{1}H} NMR (100 MHz, DMSO-d_6) δ 203.2, 184.8, 149.6, 139.5, 137.9, 137.4, 136.0, 131.7, 131.0, 128.1, 127.1, 126.8, 126.5, 125.0, 123.1, 122.0, 120.6, 114.4, 111.6, 105.2, 20.9, 19.6. HRMS (ESI) m/z calcd for $C_{27}H_{24}NO$ $[M + H]^+$: 378.1852, found: 378.1857.

*2-(4-Isopropyl-2-(3-methyl-1-phenylbuta-1,2-dien-1-yl)phenyl)-1H-indole-3-carbaldehyde (**3d**).* Yellow solid, mp = 186.6−187.3 °C, 52.7 mg (65%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 9.85 (s, 1H), 8.44 (s, 1H), 8.31 (d, J = 8.0 Hz, 1H), 7.47 (d, J = 7.6 Hz, 1H), 7.32 (d, J = 8.0 Hz, 2H), 7.24−7.11 (m, 8H), 3.06−2.95 (m, 1H), 1.48 (s, 6H), 1.32 (d, J = 6.8 Hz, 6H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 203.3, 187.0, 151.2, 149.5, 138.2, 135.3, 132.8, 129.7, 128.6, 127.0, 125.6, 124.0, 122.9, 122.1, 116.0, 110.8, 105.1, 99.7, 34.1, 24.0, 20.0. HRMS (ESI) m/z calcd for $C_{29}H_{28}NO$ $[M + H]^+$: 406.2165, found: 406.2171.

*2-(4-(tert-Butyl)-2-(3-methyl-1-phenylbuta-1,2-dien-1-yl)-phenyl)-1H-indole-3-carbaldehyde (**3e**).* Yellow solid, mp = 211−212 °C, 39.4 mg (47%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.07 (s, 1H), 9.58 (s, 1H), 8.01 (d, J = 7.2 Hz, 1H), 7.57 (dd, J = 8.0, 2.0 Hz, 1H), 7.52 (d, J = 8.0 Hz, 1H), 7.40 (d, J = 1.6 Hz, 1H), 7.34 (d, J = 7.6 Hz, 1H), 7.19−7.13 (m, 4H), 7.09−7.04 (m, 3H), 1.44 (s, 6H), 1.33 (s, 9H). ^{13}C{^{1}H} NMR (100 MHz, DMSO-d_6) δ 203.2, 184.7, 152.3, 149.5, 137.5, 137.3, 136.0, 131.6, 128.1, 127.2, 127.2, 126.7, 126.5, 125.0, 124.5, 123.1, 122.0, 120.6, 114.3, 111.6, 105.5, 98.7, 34.5, 30.9, 19.6. HRMS (ESI) m/z calcd for $C_{30}H_{29}NONa$ $[M + Na]^+$: 442.2147, found: 442.2151.

*2-(4-Chloro-2-(3-methyl-1-phenylbuta-1,2-dien-1-yl)phenyl)-1H-indole-3-carbaldehyde (**3f**).* Yellow solid, mp = 202.8−203.1 °C, 43.8 mg (55%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.16 (s, 1H), 9.58 (s, 1H), 8.00 (d, J = 7.2 Hz, 1H), 7.64−7.59 (m, 2H), 7.42 (d, J = 1.9 Hz, 1H), 7.35 (d, J = 7.6 Hz, 1H), 7.20−7.13 (m, 4H), 7.09−7.01 (m, 3H), 1.46 (s, 6H). ^{13}C{^{1}H} NMR (100 MHz, DMSO-d_6) δ 203.4, 184.6, 147.7, 140.2, 136.7, 136.0, 134.6, 133.6, 129.9, 128.8, 128.2, 127.4, 126.8, 123.3, 122.2, 120.6, 114.6, 104.5, 99.6, 19.5. HRMS (ESI) m/z calcd for $C_{26}H_{21}NOCl$ $[M + H]^+$: 398.1306, found: 398.1313.

*2-(4-Bromo-2-(3-methyl-1-phenylbuta-1,2-dien-1-yl)phenyl)-1H-indole-3-carbaldehyde (**3g**).* Yellow solid, mp = 204.6−205.0 °C, 49.5 mg (56%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.14 (s, 1H), 9.59 (s, 1H), 8.00 (d, J = 7.2 Hz, 1H), 7.74 (dd, J = 8.0, 2.0 Hz, 1H), 7.55 (dd, J = 5.6, 3.6 Hz, 2H), 7.35 (d, J = 7.6 Hz, 1H), 7.20−7.12 (m, 4H), 7.08 (d, J = 7.6 Hz, 1H), 7.02 (d, J = 7.6 Hz, 2H), 1.46 (s, 6H). ^{13}C{^{1}H} NMR (100 MHz, DMSO-d_6) δ 203.4, 184.6, 147.8, 140.4, 136.7, 136.0, 132.8, 129.2, 128.2, 126.8, 124.9, 123.3, 123.2, 122.2, 114.5, 111.7, 104.4, 99.6, 19.5. HRMS (ESI) m/z calcd for $C_{26}H_{21}NOBr$ $[M + H]^+$: 442.0801, found: 442.0806.

*2-(4-Methoxy-2-(3-methyl-1-phenylbuta-1,2-dien-1-yl)phenyl)-1H-indole-3-carbaldehyde (**3h**).* Yellow solid, mp = 205.7−206.1 °C, 48.0 mg (61%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.02 (s, 1H), 9.58 (s, 1H), 8.00 (d, J = 7.2 Hz, 1H), 7.53 (d, J = 8.4 Hz, 1H), 7.33 (d, J = 7.2 Hz, 1H), 7.17 (dd, J = 6.8, 1.2 Hz, 2H), 7.15−7.10 (m, 3H), 7.08−7.05 (m, 3H), 6.91 (d, J = 2.8 Hz, 1H), 3.83 (s, 3H), 1.46 (s, 6H). ^{13}C{^{1}H} NMR (100 MHz, DMSO-d_6) δ 203.3, 185.1, 160.3, 149.6, 139.7, 137.2, 136.1, 133.3, 128.3, 127.0, 126.7, 125.3, 123.2, 122.3, 120.7, 116.2, 114.6, 113.0, 111.8, 105.5, 99.0, 55.6, 19.8. HRMS (ESI) m/z calcd for $C_{27}H_{24}NO_2$ $[M + H]^+$: 394.1802, found: 394.1807.

*Methyl-4-(3-formyl-1H-indol-2-yl)-3-(3-methyl-1-phenylbuta-1,2-dien-1-yl)benzoate (**3i**).* Yellow solid, mp = 190.8−191.2 °C, 54.0 mg (64%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.25 (s, 1H), 9.61 (s, 1H), 8.06 (dd, J = 8.0, 1.6 Hz, 1H), 8.03 (d, J = 7.7 Hz, 1H), 7.92 (d, J = 1.4 Hz, 1H), 7.76 (d, J = 8.0 Hz, 1H), 7.38 (d, J = 7.6 Hz, 1H), 7.22−7.14 (m, 4H), 7.13−7.09 (m, 1H), 7.08−7.04 (m, 2H), 3.88 (s, 3H), 1.41 (s, 6H). ^{13}C{^{1}H} NMR (100 MHz, DMSO-d_6) δ 203.6, 184.7, 165.8, 147.9, 137.1, 136.3, 134.8, 132.8, 128.5, 127.9, 127.0, 123.6, 122.4, 120.9, 114.7, 112.0, 104.8, 99.7, 52.7, 19.7. HRMS (ESI) m/z calcd for $C_{28}H_{24}NO_3$ $[M + H]^+$: 422.1751, found: 422.1756.

*2-(2-Methyl-6-(3-methyl-1-phenyl-2l5-buta-1,2-dien-1-yl)-phenyl)-1H-indole-3-carbaldehyde (**3j**).* Yellow solid, mp = 229−231 °C, 15.1 mg (20%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.08 (s, 1H), 9.44 (s, 1H), 8.02 (d, J = 7.2 Hz, 1H), 7.49 (t, J = 7.6 Hz, 1H), 7.40 (d, J = 7.6 Hz, 1H), 7.33 (d, J = 7.2 Hz, 1H), 7.23−7.15(m, 5H), 7.08 (t, J = 7.2 Hz, 1H), 6.99 (d, J = 7.2 Hz, 2H), 2.11 (s, 3H), 1.57 (s, 3H), 1.29 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, DMSO-d_6) δ 202.5, 184.2, 148.3, 138.7, 138.4, 137.6, 136.1, 129.7, 129.5, 129.0, 128.1, 127.7, 126.7, 126.4, 124.8, 123.1, 122.0, 120.7, 114.3, 111.7, 105.0, 98.5, 19.9. HRMS (ESI) m/z calcd for $C_{27}H_{23}NONa$ $[M + Na]^+$: 400.1677, found: 400.1679.

*2-(2-Chloro-6-(3-methyl-1-phenylbuta-1,2-dien-1-yl)phenyl)-1H-indole-3-carbaldehyde (**3k**).* Yellow solid, mp = 180.9−181.9 °C, 22.2 mg (28%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz,

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$CDCl_3$) δ 9.64 (s, 1H), 8.39 (s, 1H), 8.27 (d, J = 6.8 Hz, 1H), 7.53−7.44 (m, 2H), 7.39 (d, J = 7.2 Hz, 1H), 7.23 (dd, J = 9.2, 2.4 Hz, 3H), 7.14−7.06(m, 3H), 6.98 (d, J = 7.2 Hz, 2H), 1.55 (s, 3H), 1.48 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.1, 186.0, 144.6, 142.0, 137.9, 135.7, 130.9, 129.3, 128.8, 128.5, 128.2, 126.9, 126.8, 124.9, 123.9, 122.8, 122.0, 116.5, 110.9, 104.7, 99.9, 20.2, 20.0. HRMS (ESI) m/z calcd for $C_{26}H_{20}NOClNa$ $[M + Na]^+$: 420.1131, found: 420.1134.

*2-(2-Bromo-6-(3-methyl-1-phenylbuta-1,2-dien-1-yl)phenyl)-1H-indole-3-carbaldehyde (**3l**).* Yellow solid, mp = 175.8−176.5 °C, 19.4 mg (22%). Purification (PE/EA = 5/1). ^1H NMR (400 MHz, $CDCl_3$) δ 9.65 (s, 1H), 8.34−8.24 (m, 2H), 7.52−7.45 (m, 2H), 7.39 (d, J = 7.2 Hz, 1H), 7.25−7.18 (m, 3H), 7.15−7.08 (m, 3H), 6.98 (d, J = 6.8 Hz, 2H), 1.55 (s, 3H), 1.48 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.2, 186.2, 144.9, 142.1, 138.0, 135.9, 135.8, 131.0, 129.4, 128.9, 128.6, 128.4, 127.0, 126.9, 125.1, 124.0, 122.9, 122.1, 116.6, 111.2, 104.9, 100.1, 20.3, 20.1. HRMS (ESI) m/z calcd for $C_{26}H_{20}NOBrNa$ $[M + Na]^+$: 464.0626, found: 464.0624.

*5-Chloro-2-(2-(3-methyl-1-phenylbuta-1,2-dien-1-yl)phenyl)-1H-indole-3-carbaldehyde (**3m**).* Yellow solid, mp = 89.8−90.2 °C. 59.7 mg (75%). Purification (PE/EA = 5/1). ^1H NMR (400 MHz, $CDCl_3$) δ 9.69 (s, 1H), 9.00 (s, 1H), 8.27 (d, J = 1.6 Hz, 1H), 7.56−7.51 (m, 1H), 7.50−7.47 (m, 2H), 7.45−7.40 (m, 1H), 7.19−7.12 (m, 4H), 7.11−7.05 (m, 3H), 1.49 (s, 6H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.5, 186.6, 150.4, 138.7, 138.0, 133.8, 131.5, 130.3, 129.1, 128.7, 128.5, 127.5, 127.0, 127.0, 126.5, 124.4, 121.5, 115.4, 112.1, 105.1, 99.7, 20.0. HRMS (ESI) m/z calcd for $C_{26}H_{21}NOCl$ $[M + H]^+$: 398.1306, found: 398.1311.

*5-Bromo-2-(2-(3-methyl-1-phenylbuta-1,2-dien-1-yl)phenyl)-1H-indole-3-carbaldehyde (**3n**).* Yellow solid, mp = 99.7−100.1 °C. 62.8 mg (71%). Purification (PE/EA = 5/1). ^1H NMR (400 MHz, DMSO-d_6) δ 12.33 (s, 1H), 9.55 (s, 1H), 8.12 (t, J = 1.2 Hz, 1H), 7.63−7.58 (m, 2H), 7.55−7.51(m, 1H), 7.45−7.42 (m, 1H), 7.31 (d, J = 1.2 Hz, 2H), 7.17−7.12 (m, 2H), 7.07−6.99 (m, 3H), 1.47 (s, 6H). $^{13}C\{^1H\}$ NMR (100 MHz, DMSO-d_6) δ 203.3, 184.7, 150.1, 138.0, 137.2, 134.8, 131.7, 130.7, 130.2, 129.3, 128.1, 127.5, 126.8, 126.7, 126.6, 125.8, 122.7, 114.7, 113.8, 113.7, 105.3, 98.8, 19.6. HRMS (ESI) m/z calcd for $C_{26}H_{21}NOBr$ $[M + H]^+$: 442.0801, found: 442.0806.

*5-Fluoro-2-(2-(3-methyl-1-phenylbuta-1,2-dien-1-yl)phenyl)-1H-indole-3-carbaldehyde (**3o**).* Yellow solid, mp = 197.6−198.6 °C, 47.3 mg (62%). Purification (PE/EA = 5/1). ^1H NMR (400 MHz, $CDCl_3$) δ 9.78 (s, 1H), 8.42 (s, 1H), 7.97 (dd, J = 9.6, 2.4 Hz, 1H), 7.57−7.51 (m, 2H), 7.51−7.43 (m, 2H), 7.22−7.17 (m, 2H), 7.14−7.10 (m, 4H), 6.98−6.93 (m, 1H), 1.49 (s, 6H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.4, 186.6, 161.0, 158.6, 150.3, 138.6, 138.1, 132.5, 131.7, 131.6, 130.3, 129.5, 129.2, 128.6, 127.6, 127.0, 126.2 (d, J = 11.3 Hz), 116.1 (d, J = 4.3 Hz), 112.4 (d, J = 26.3 Hz), 111.6 (d, J = 9.6 Hz), 107.6 (d, J = 24.9 Hz), 105.0, 99.8, 20.0. ^{19}F NMR (565 MHz, $CDCl_3$) δ −121.37 to −121.82 (m, 1F). HRMS (ESI) m/z calcd for $C_{26}H_{20}NOFNa$ $[M + Na]^+$: 404.1427, found: 404.1431.

*2-(2-(3-Methyl-1-phenylbuta-1,2-dien-1-yl)phenyl)-5-nitro-1H-indole-3-carbaldehyde (**3p**).* Yellow solid, mp = 184.6−185.0 °C, 58.8 mg (72%). Purification (PE/EA = 5/1). ^1H NMR (400 MHz, DMSO-d_6) δ 12.78 (s, 1H), 9.64 (s, 1H), 8.84 (s, 1H), 8.09 (dd, J = 8.8, 2.4 Hz, 1H), 7.65 (t, J = 6.8 Hz, 2H), 7.61−7.56 (m, 1H), 7.53 (d, J = 8.8 Hz, 1H), 7.49 (d, J = 7.6 Hz, 1H), 7.12 (t, J = 7.2 Hz, 2H), 7.03 (d, J = 7.2 Hz, 1H), 7.01−6.97 (m, 2H), 1.51 (s, 6H).$^{13}C\{^1H\}$ NMR (100 MHz, DMSO-d_6) δ 203.5, 185.0, 152.2, 142.8, 139.2, 138.2, 131.6, 130.8, 130.5, 128.6, 128.1, 127.7, 126.8, 126.6, 124.4, 116.7, 115.2, 105.4, 98.9, 19.6. HRMS (ESI) m/z calcd for $C_{26}H_{21}N_2O_3$ $[M + H]^+$: 409.1547, found: 409.1552.

*6-Methyl-2-(2-(3-methyl-1-phenybuta-1,2-dien-1-yl)phenyl)-1H-indole-3-carbaldehyde (**3q**).* Yellow solid, mp = 201.9−203.9 °C, 38.5 mg (51%). Purification (PE/EA = 5/1). ^1H NMR (400 MHz, DMSO-d_6) δ 12.00 (s, 1H), 9.55 (s, 1H), 7.92−7.82 (m, 1H), 7.61−7.51 (m, 3H), 7.42 (d, J = 7.6 Hz, 1H), 7.25−7.13 (m, 3H), 7.06 (dd, J = 15.6, 7.2 Hz, 3H), 7.02−6.96 (m, 1H), 2.37 (s, 3H), 1.46 (s, 6H). $^{13}C\{^1H\}$ NMR (100 MHz, DMSO-d_6) δ 203.2, 184.6, 148.9, 138.0, 137.3, 136.4, 131.8, 130.6, 130.1, 129.8, 128.1, 127.4, 126.8, 126.5, 123.7, 122.8, 120.3, 114.3, 111.5, 105.2, 98.8, 21.3, 19.6. HRMS (ESI) m/z calcd for $C_{27}H_{23}NONa$ $[M + H]^+$: 400.1667, found: 400.1669.

*2-(2-(3-Methyl-1-phenylpenta-1,2-dien-1-yl)phenyl)-1H-indole-3-carbaldehyde (**3r**).* Yellow solid, mp = 193.3−193.5 °C, 46.4 mg (61%). Purification (PE/EA = 5/1). ^1H NMR (400 MHz, $CDCl_3$) δ 9.79 (s, 1H), 8.80 (s, 1H), 8.32−8.26 (m, 1H), 7.53 (d, J = 7.6 Hz, 3H), 7.44 (t, J = 7.2 Hz, 1H), 7.25−7.17 (m, 4H), 7.16−7.08 (m, 4H), 1.92 (q, J = 7.6 Hz, 2H), 1.44 (s, 3H), 0.92 (t, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 202.5, 186.8, 149.3, 138.6, 138.2, 135.4, 132. 7, 131. 4, 130.1, 129.5, 128.5, 127.4, 126.9, 126.8, 125.4, 124.0, 122.9, 122.0, 115.9, 110.9, 106.9, 106.0, 27.3, 18.3, 12.3. HRMS (ESI) m/z calcd for $C_{27}H_{23}NONa$ $[M + Na]^+$: 400.1677, found 400.1680.

5 mmol-Scale Experiment of 3m. A mixture of **1l** (5 mmol, 1278 mg), **2a** (7.5 mmol, 1952 mg), $[Cp^*RhCl_2]_2$ (2.5 mol %, 77.26 mg), AgOAc (0.4 mmol, 2 equiv, 1669 mg), and EtOAc (2 mL) in a 50 mL sealed tube was heated in an oil bath at 60 °C for 24 h under an air atmosphere. The reaction mixture was cooled to room temperature and concentrated under reduced pressure, and then the residue was subjected to silica gel column chromatography [eluting with petroleum ether (PE)/ethyl acetate (EA)] to afford the corresponding product **3m** (1.11 g, 56% yield).

Procedure for the Synthesis of 3aa. Under an air atmosphere, **3a** (0.3 mmol, 109 mg), $NaClO_2$ (1.5 mmol, 135.7 mg), $NaH_2PO_4 \cdot 2H_2O$ (4.5 mmol, 702 mg), and 2-methyl-2-butene (550 μL) were dissolved in t-BuOH (600 μL), THF (600 μL), and H_2O (600 μL). The reaction was performed at room temperature for 24 h. Upon completion, the solvent was removed under reduced pressure, and the residue was diluted with EtOAc (5 mL) and extracted with H_2O (3 × 10 mL). The organic layers were combined and dried over anhydrous Na_2SO_4 and concentrated under vacuum. The residue was subjected to silica gel column chromatography [eluting with petroleum ether (PE)/ethyl acetate (EA)] to afford the corresponding product **3aa**.

*2-(2-(3-Methyl-1-phenyl-1,2-dien-1-yl)phenyl)-1H-indole-3-carboxylic Acid (**3aa**).* White solid, mp = 193.4−194.1 °C, 51.2 mg (45%). Purification (PE/EA = 3/1). ^1H NMR (400 MHz, DMSO-d_6) δ 11.80 (s, 1H), 11.57 (s, 1H), 7.98 (d, J = 7.4 Hz, 1H), 7.42 (d, J = 4.0 Hz, 3H), 7.32 (d, J = 7.6 Hz, 1H), 7.21 (dd, J = 10.2, 7.6 Hz, 5H), 7.11 (m, 3H), 1.32 (s, 6H). $^{13}C\{^1H\}$ NMR (100 MHz, DMSO-d_6) δ 202.8, 165.9, 144.4, 138.3, 137.7, 135.6, 133.4, 131.3, 129.4, 128.5, 127.9, 127.7, 127.3, 126.5, 126.4, 121.8, 121.0, 120.8, 111.4, 105.1, 104.5, 98.1, 20.0. HRMS (ESI) m/z calcd for $C_{26}H_{22}NO_2$ $[M + H]^+$: 380.1651, found: 380.1647.

Procedure for the Synthesis of 3ab. **3a** (0.2 mmol, 72.7 mg) and $NaBH_4$(0.4 mmol, 2 equiv) were dissolved in MeOH (500 μL) and THF (500 μL). The reaction was performed at room temperature for 1 h. After completion of the reaction, the resulting mixture was quenched with H_2O (10 mL), and the filtrate was filtered and dried to afford the corresponding product **3ab**.

*(2-(2-(3-Methyl-1-phenyl-1,2-dien-1-yl)phenyl)-1H-indol-3-yl)-methanol (**3ab**).* White solid, mp = 145.6−146.1 °C, 65.1 mg (89%). ^1H NMR (400 MHz, $CDCl_3$) δ 8.10 (s, 1H), 7.73 (d, J = 7.2 Hz, 1H), 7.59 (d, J = 6.8 Hz, 1H), 7.52−7.46 (m, 3H), 7.29 (d, J = 10.8 Hz, 1H), 7.24−7.14 (m, 7H), 4.76 (d, J = 5.6 Hz, 2H), 1.56 (s, 6H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 202.9, 138.3, 137.9, 136.3, 135.6, 132.0, 131.8, 131.2, 128.8, 128.4, 127.7, 127.5, 126.9, 126.8, 122.2, 119.9, 119.0, 113.0, 110.6, 105.4, 99.2, 56.3, 20.0. HRMS (ESI) m/z calcd for $C_{26}H_{23}NONa$ $[M + H]^+$: 388.1677, found 388.1671.

General Procedure for the Synthesis of 4a−4p. A mixture of **3** (0.2 mmol), EtONa (0.4 mmol, 2 equiv, 13.6 mg), and DMF (2 mL) in a 50 mL Schlenk tube was stirred at 60 °C in an oil bath for 5 h under an Ar atmosphere. The resulting mixture was cooled to room temperature and concentrated, and the resulting mixture was subjected to silica gel column chromatography [eluting with petroleum ether (PE)/ethyl acetate (EA)] to afford the corresponding product (**4a−4p**).

*5-Phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo[2,1-a]-isoquinoline-12-carbaldehyde (**4a**).* White solid, mp = 204.2−204.6 °C, 30.1 mg (83%). Purification (PE/EA = 4/1). ^1H NMR (400 MHz, $CDCl_3$) δ 10.48 (s, 1H), 8.38 (d, J = 7.6 Hz, 1H), 8.06−8.04

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(m,1H), 7.51 (d, J = 2.4 Hz, 3H), 7.31−7.20 (m, 2H), 7.15 (d, J = 8.0 Hz, 1H), 7.05−6.97 (m, 3H), 6.90 (d, J = 7.2 Hz, 2H), 5.35 (s, 1H), 2.20 (s, 3H), 1.78 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 186.1, 143.6, 139.6, 138.9, 136.2, 130.1, 129.4, 128.8, 128.6, 128.2, 127.3, 127.1, 127.0, 126.8, 126.5, 124.2, 123.1, 122.1, 113.6, 112.2, 47.9, 23.0, 20.4. HRMS (ESI) m/z calcd for $C_{26}H_{22}NO$ $[M + H]^+$: 364.1701, found: 364.1700. It was crystallized from PE/EA.

*4-Methyl-5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo[2,1-a]isoquinoline-12-carbaldehyde (**4b**).* White solid, mp = 206.0−206.6 °C, 41.7 mg (97%). Purification (PE/EA = 5/1). ^1H NMR (400 MHz, $CDCl_3$) δ 10.48 (s, 1H), 8.37 (d, J = 8.4 Hz, 1H), 7.85 (s, 1H), 7.41 (d, J = 7.6 Hz, 1H), 7.34 (d, J = 6.8 Hz, 1H), 7.30−7.20 (m, 2H), 7.14 (d, J = 7.6 Hz, 1H), 7.06−6.97 (m, 3H), 6.90 (d, J = 6.8 Hz, 2H), 5.32 (s, 1H), 2.50 (s, 3H), 2.20 (s, 3H), 1.77 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 186.3, 144.0, 139.8, 138.1, 136.3, 136.0, 130.9, 129.4, 129.3, 128.5, 127.6, 126.9, 126.7, 126.4, 126.3, 124.1, 123.1, 122.2, 113.4, 112.2, 47.4, 22.9, 21.5, 20.4. HRMS (ESI) m/z calcd for $C_{27}H_{23}NONa$ $[M + Na]^+$: 400.1677, found: 400.1684.

*3-Methoxy-5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo-[2,1-a]isoquinoline-12-carbaldehyde (**4c**).* White solid, mp = 167.3−167.9 °C, 35.4 mg (90%). Purification (PE/EA = 3/1). ^1H NMR (400 MHz, $CDCl_3$) δ 10.41 (s, 1H), 8.32 (dd, J = 6.8, 1.2 Hz, 1H), 8.04−8.00 (m, 1H), 7.24−7.18 (m, 2H), 7.09 (dd, J = 7.2, 1.6 Hz, 1H), 7.06−7.00 (m, 5H), 6.91 (d, J = 6.4 Hz, 2H), 5.29 (s, 1H), 3.91 (s, 3H), 2.20 (s, 3H), 1.77 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 185.9, 161.2, 144.1, 140.8, 139.5, 136.1, 130.4, 128.6, 127.4, 127.1, 126.8, 126.7, 126.4, 123.8, 123.0, 121.8, 119.9, 115.2, 113.4, 112.8, 112.1, 55.7, 48.2, 23.0, 20.4. HRMS (ESI) m/z calcd for $C_{27}H_{23}NO_2Na$ $[M + Na]^+$: 416.1626, found 416.1631.

*Methyl 12-Formyl-5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo[2,1-a]isoquinoline-3-carboxylate (**4d**).* Yellow solid, mp = 229.3−229.9 °C, 40.0 mg (95%). Purification (PE/EA = 10/1). ^1H NMR (400 MHz, $CDCl_3$) δ 10.48 (s, 1H), 8.35 (d, J = 7.6 Hz, 1H), 8.21−8.11 (m, 3H), 7.33−7.21 (m, 2H), 7.16 (d, J = 8.4 Hz, 1H), 7.06−6.97 (m, 3H), 6.89−6.85 (m, 2H), 5.42 (s, 1H), 3.98 (s, 3H), 2.21 (s, 3H), 1.78 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 185.6, 166.3, 141.6, 138.9, 138.9, 136.2, 131.0, 130.8, 130.2, 129.2, 128.5, 128.5, 127.1, 127.0, 126.6, 126.5, 126.3, 124.6, 123.3, 122.1, 114.1, 112.2, 52.5, 47.6, 22.9, 20.3. HRMS (ESI) m/z calcd for $C_{28}H_{23}NO_3Na$ $[M + Na]^+$: 444.1576, found 444.1573.

*3-Bromo-5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo[2,1-a]isoquinoline-12-carbaldehyde (**4e**).* White solid, mp = 159.6−160.2 °C, 30.6 mg (69%). Purification (PE/EA = 10/1). ^1H NMR (400 MHz, $CDCl_3$) δ 10.43 (s, 1H), 8.32 (d, J = 8.4 Hz, 1H), 7.97 (d, J = 8.4 Hz, 1H), 7.68−7.63 (m, 2H), 7.30−7.21 (m, 2H), 7.12 (d, J = 8.0 Hz, 1H), 7.07−6.98 (m, 3H), 6.90−6.85 (m, 2H), 5.30 (s, 1H), 2.20 (s, 3H), 1.78 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 185.4, 142.0, 140.5, 138.8, 136.1, 132.1, 131.3, 130.0, 128.6, 127.1, 126.5, 126.4, 125.9, 124.3, 124.2, 123.1, 121.8, 113.4, 112.1, 47.5, 22.9, 20.3. HRMS (ESI) m/z calcd for $C_{26}H_{20}NOBrNa$ $[M + Na]^+$: 464.0626, found: 464.0635.

*3-Chloro-5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo[2,1-a]isoquinoline-12-carbaldehyde (**4f**).* White solid, mp = 196.1−196.7 °C, 37 mg (93%). Purification (PE/EA = 10/1). ^1H NMR (400 MHz, $CDCl_3$) δ 10.43 (s, 1H), 8.32 (d, J = 7.2 Hz, 1H), 8.04 (d, J = 8.4 Hz, 1H), 7.50 (d, J = 7.6 Hz, 2H), 7.31−7.26 (m, 1H), 7.25−7.21 (m, 1H), 7.12 (d, J = 8.0 Hz, 1H), 7.07−6.99 (m, 3H), 6.89−6.85 (m, 2H), 5.31 (s, 1H), 2.20 (s, 3H), 1.78 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 185.6, 142.2, 140.5, 138.9, 136.2, 136.0, 130.0, 129.4, 128.7, 128.5, 127.3, 127.2, 126.7, 126.6, 125.7, 124.4, 123.3, 122.0, 113.6, 112.3, 47.7, 23.0, 20.5. HRMS (ESI) m/z calcd for $C_{26}H_{20}NOClNa$ $[M + Na]^+$: 420.1131, found: 420.1135.

*3-Methyl-5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo[2,1-a]isoquinoline-12-carbaldehyde (**4g**).* White solid, mp = 205.9−206.4 °C, 34.7 mg (92%). Purification (PE/EA = 5/1). ^1H NMR (400 MHz, $CDCl_3$) δ 10.46 (s, 1H), 8.36 (d, J = 7.2 Hz, 1H), 7.94 (d, J = 8.4 Hz, 1H), 7.33 (d, J = 6.0 Hz, 2H), 7.28−7.20 (m, 2H), 7.12 (d, J = 8.0 Hz, 1H), 7.07−6.97 (m, 3H), 6.91 (d, J = 7.2 Hz, 2H), 5.30 (s, 1H), 2.48 (s, 3H), 2.20 (s, 3H), 1.77 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 186.1, 144.1, 140.6, 139.7, 138.8, 136.2, 130.1, 129.0, 128.7, 128.6, 127.5, 127.0, 126.8, 126.5, 126.3, 124.3, 124.0, 123.0, 122.0, 113.2, 112.2, 47.9, 23.0, 21.7, 20.4. HRMS (ESI) m/z calcd for $C_{27}H_{23}NONa$ $[M + Na]^+$: 400.1677, found: 400.1682.

*3-Isopropyl-5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo-[2,1-a]isoquinoline-12-carbaldehyde (**4h**).* White solid, mp = 159.9−162.2 °C, 33.6 mg (83%). Purification (PE/EA = 5/1). ^1H NMR (400 MHz, $CDCl_3$) δ 10.47 (s, 1H), 8.36 (d, J = 7.6 Hz, 1H), 7.98 (d, J = 7.6 Hz, 1H), 7.41−7.35 (m, 2H), 7.29−7.20 (m, 2H), 7.12 (d, J = 7.6 Hz, 1H), 7.07−6.97 (m, 3H), 6.90 (d, J = 7.2 Hz, 2H), 5.33 (s, 1H), 3.09−2.98 (m,1H), 2.21 (s, 3H), 1.77 (s, 3H), 1.36 (dd, J = 6.8, 4.0 Hz, 6H).$^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 186.1, 151.5, 144.0, 139.8, 138.9, 136.2, 128.8, 128.6, 127.6, 127.0, 126.8, 126.5, 126.4, 126.2, 124.7, 124.0, 123.0, 122.0, 113.3, 112.2, 48.0, 34.3, 23.9, 23.0, 20.5. HRMS (ESI) m/z calcd for $C_{29}H_{27}NONa$ $[M + Na]^+$: 428.1990, found: 428.1995.

*3-(tert-Butyl)-5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo-[2,1-a]isoquinoline-12-carbaldehyde (**4i**).* White solid, mp = 197.1−197.6 °C, 35 mg (83%). Purification (PE/EA = 5/1). ^1H NMR (400 MHz, $CDCl_3$) δ 10.48 (s, 1H), 8.36 (dd, J = 6.8, 1.2 Hz, 1H), 8.00 (d, J = 8.0 Hz, 1H), 7.58−7.48 (m, 2H), 7.28−7.20 (m,2H), 7.12 (d, J = 7.2 Hz, 1H), 7.08−6.98 (m, 3H), 6.91 (d, J = 7.2 Hz, 2H), 5.34 (s, 1H), 2.22 (s, 3H), 1.77 (s, 3H), 1.43 (s, 9H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 186.2, 153.8, 143.9, 139.8, 138.5, 136.2, 128.6, 128.6, 128.5, 127.6, 126.9, 126.8, 126.5, 126.4, 126.2, 125.4, 124.4, 124.0, 123.0, 122.0, 113.3, 112.2, 48.1, 35.2, 31.3, 23.0, 20.5. HRMS (ESI) m/z calcd for $C_{30}H_{29}NONa$ $[M + Na]^+$: 442.2147, found: 442.2150.

*1-Bromo-5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo[2,1-a]isoquinoline-12-carbaldehyde (**4j**).* White solid, mp = 166.1−166.7 °C, 38 mg (86%). Purification (PE/EA = 10/1). ^1H NMR (400 MHz, $CDCl_3$) δ 10.36 (s, 1H), 8.31 (d, J = 7.2 Hz, 1H), 7.57 (dd, J = 6.4, 2.8 Hz, 1H), 7.45−7.40 (m, 2H), 7.27−7.17 (m, 2H), 7.13 (d, J = 7.2 Hz, 1H), 6.97−6.94 (m, 3H), 6.84−6.79 (m, 2H), 5.29 (s, 1H), 2.18 (s, 3H), 1.91 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 187.2, 142.1, 139.9, 138.4, 137.2, 132.9, 130.7, 130.6, 128.5, 127.6, 127.2, 127.1, 126.9, 126.6, 126.2, 124.3, 123.2, 122.5, 113.9, 112.0, 49.0, 22.8, 20.2. HRMS (ESI) m/z calcd for $C_{26}H_{20}NOBrNa$ $[M + Na]^+$: 464.0626, found: 464.0619.

*1-Chloro-5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo[2,1-a]isoquinoline-12-carbaldehyde (**4k**).* White solid, mp = 177.9−178.3 °C, 35.8 mg (88%). Purification (PE/EA = 5/1). ^1H NMR (400 MHz, $CDCl_3$) δ 10.37 (d, J = 0.8 Hz, 1H), 8.34−8.28 (m, 1H), 7.59−7.55 (m, 1H), 7.45−7.39 (m, 2H), 7.28−7.17 (m, 2H), 7.13 (d, J = 8.0 Hz, 1H), 7.00−6.92 (m, 3H), 6.82 (dd, J = 8.4, 2.4 Hz, 2H), 5.29 (s, 1H), 2.18 (s, 3H), 1.91 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 187.1, 142.0, 139.7, 138.2, 137.1, 132.7, 130.6, 130.5, 128.3, 127.5, 127.0, 127.0, 126.8, 126.5, 126.0, 124.2, 123.0, 122.3, 113.7, 111.8, 48.9, 22.7, 20.1. HRMS (ESI) m/z calcd for $C_{26}H_{20}NOClNa$ $[M + Na]^+$: 420.1131, found: 420.1135.

*1-Methyl-5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo[2,1-a]isoquinoline-12-carbaldehyde (**4l**).* White solid, mp = 159.7−160.0 °C, 21.4 mg (55%). Purification (PE/EA = 5/1). ^1H NMR (400 MHz, $CDCl_3$) δ 10.04 (d, J = 2.0 Hz, 1H), 8.28 (d, J = 7.6 Hz, 1H), 7.42 (dd, J = 5.6, 2.4 Hz, 2H), 7.33 (dd, J = 5.2, 2.4 Hz, 1H), 7.27−7.17 (m, 2H), 7.14 (d, J = 8.0 Hz, 1H), 6.98−6.90 (m, 3H), 6.84 (d, J = 7.6 Hz, 2H), 5.27 (s, 1H), 2.71 (d, J = 2.0 Hz, 3H), 2.19 (d, J = 2.0 Hz, 3H), 1.91 (d, J = 2.0 Hz, 3H).$^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 186.7, 143.6, 140.3, 139.0, 137.3, 137.0, 131.3, 129.7, 128.3, 127.8, 127.3, 126.9, 126.7, 126.5, 126.4, 126.2, 124.0, 123.0, 122.0, 113.7, 112.0, 22.8, 21.5, 20.2. HRMS (ESI) m/z calcd for $C_{27}H_{23}NONa$ $[M + Na]^+$: 400.1677, found: 400.1684.

*10-Bromo-5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo[2,1-a]isoquinoline-12-carbaldehyde (**4m**).* White solid, mp = 225.8−226.4 °C, 39 mg (88%). Purification (PE/EA = 5/1). ^1H NMR (400 MHz, $CDCl_3$) δ 10.41 (s, 1H), 8.53 (d, J = 1.6 Hz, 1H), 8.00 (d, J = 7.6 Hz, 1H), 7.53 (d, J = 2.4 Hz, 3H), 7.35 (dd, J = 8.8, 2.0 Hz, 1H), 7.11−6.97 (m, 4H), 6.87 (d, J = 6.0 Hz, 2H), 5.35 (s, 1H), 2.20 (s, 3H), 1.76 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 185.8, 144.3, 139.4, 138.8, 134.9, 130.5, 129.5, 128.7, 128.7, 128.4, 127.9, 127.2, 126.9, 126.7, 124.8, 116.6, 113.5, 112.9, 47.7, 22.9, 20.4. HRMS (ESI)

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m/z calcd for $C_{26}H_{20}NOBrNa$ $[M + Na]^+$: 464.0626, found: 464.0632.

*10-Chloro-5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo[2,1-a]isoquinoline-12-carbaldehyde (**4n**).* White solid, mp = 227.1−227.7 °C, 37.9 mg (97%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.43 (s, 1H), 8.37 (d, J = 1.6 Hz, 1H), 8.06−7.96 (m, 1H), 7.53 (d, J = 2.8 Hz, 3H), 7.22 (dd, J = 8.8, 2.0 Hz, 1H), 7.06−7.02 (m, 4H), 6.87 (d, J = 6.8 Hz, 2H), 5.35 (s, 1H), 2.20 (s, 3H), 1.76 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 185.7, 144.2, 139.2, 138.7, 134.5, 130.3, 129.4, 128.7, 128.6, 128.5, 128.2, 127.2, 127.1, 127.0, 126.7, 126.6, 126.5, 124.4, 121.7, 112.9, 112.8, 47.6, 22.7, 20.2. HRMS (ESI) m/z calcd for $C_{26}H_{20}NOClNa$ $[M + Na]^+$: 420.1131, found: 420.1136.

*10-Fluoro-5-phenyl-6-(propan-2-ylidene)-5,6-dihydroindolo[2,1-a]isoquinoline-12-carbaldehyde (**4o**).* White solid, mp = 204.8−205.4 °C, 36.8 mg (95%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.43 (s, 1H), 8.13−7.94 (m, 2H), 7.56−7.48 (m, 3H), 7.08−7.03 (m, 2H), 7.03−6.96 (m, 3H), 6.90−6.85 (m, 2H), 5.34 (s, 1H), 2.20 (s, 3H), 1.78 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 185.7, 160.9, 158.5, 144.4, 139.3, 138.6, 132.6, 130.2, 129.4, 128.5 (d, J = 5.7 Hz), 128.1, 127.2, 126.9 (d, J = 8.5 Hz), 126.7, 126.6 (d, J = 3.9 Hz), 113.3 (d, J = 4.2 Hz), 112.8 (d, J = 9.5 Hz), 112.3 (d, J = 26.3 Hz), 107.6 (d, J = 25.1 Hz), 47.6, 22.8, 20.2.HRMS (ESI) m/z calcd for $C_{26}H_{20}NOFNa$ $[M + Na]^+$: 404.1427, found: 404.1433.

*E-6-(Butan-2-ylidene)-5-phenyl-5,6-dihydroindolo[2,1-a]-isoquinoline-12-carbaldehyde (**4p**) and Z-6-(Butan-2-ylidene)-5-phenyl-5,6-dihydroindolo[2,1-a]isoquinoline-12-carbaldehyde (**4p′**).* Yellow solid (58:42 mixture of isomer), 62.2 mg (82%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, $CDCl_3$) 10.48 (s, 1H × 58/100), 10.47 (s, 1H × 42/100), 8.37 (d, J = 7.6 Hz, 1H), 8.05 (d, J = 3.6 Hz, 1H), 7.55−7.49 (m, 3H), 7.32−7.20 (m, 2H), 7.16 (t, J = 8.4 Hz, 1H), 7.06−6.96 (m, 3H), 6.94−6.87 (m, 2H), 5.34 (s, 1H × 58/100), 5.33 (s, 1H × 42/100), 2.66−2.49 (m, 1H), 2.33−2.22 (m, 1H × 58/100), 2.20 (s, 3H × 42/100), 2.18−2.10 (m, 1H × 42/100), 1.78 (s, 3H × 58/100), 1.25 (t, J = 7.6 Hz, 3H × 58/100), 0.93 (t, J = 7.6 Hz, 3H × 42/100). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 186.1, 186.1, 143.9, 143.6, 139.5, 139.2, 139.0, 136.4, 136.2, 132.4, 131.7, 130.1, 130.1, 129.4, 129.2, 128.9, 128.7, 128.6, 128.2, 128.2, 127.1, 127.1, 127.0, 126.8, 126.5, 126.5, 124.2, 124.1, 123.2, 123.1, 122.2, 113.6, 113.5, 112.2, 112.0, 48.1, 47.7, 28.4, 27.4, 20.2, 17.3, 13.7, 12.1. HRMS (ESI) m/z calcd for $C_{27}H_{23}NONa$ $[M + Na]^+$: 400.1677, found 400.1681.

General Procedure for the Synthesis of 5a−5m. A mixture of **3** (0.2 mmol), $Pd(OAc)_2$ (7.3 mg, 15 mol %), and cyclohexane (2 mL) in a 20 mL Schlenk tube was stirred at 100 °C in an oil bath for 24 h under an Ar atmosphere. The resulting mixture was cooled to room temperature and concentrated under reduced pressure, and the residue was subjected to silica gel column chromatography [eluting with petroleum ether (PE)/ether acetate (EA)] to afford the corresponding product (**5a−5m**).

*6-(2-Methylprop-1-en-1-yl)-6-phenyl-6H-isoindolo[2,1-a] indole-11-carbaldehyde (**5a**).* Yellow solid, mp = 79.6−81.4 °C, 54.5 mg (75%). Purification (PE/EA = 20/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.63 (s, 1H), 8.33 (d, J = 8.4 Hz, 2H), 7.47−7.35 (m, 3H), 7.33−7.26 (m, 2H), 7.25−7.17 (m, 3H), 7.13−7.08 (m, 3H), 6.34 (s, 1H), 1.80 (s, 3H), 0.77 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 183.3, 153.6, 149.5, 142.6, 142.5, 132.9, 130.8, 130.2, 129.3, 129.1, 128.6, 128.0, 125.5, 124.2, 123.8, 123.6, 122.9, 122.8, 121.6, 112.0, 110.2, 72.3, 27.8, 17.7. HRMS (ESI) m/z calcd for $C_{26}H_{22}NO$ $[M + H]^+$: 364.1701, found: 364.1707.

*Methyl 11-Formyl-6-(2-methylprop-1-en-1-yl)-6-phenyl-6H-isoindolo[2,1-a]indole-8-carboxylate (**5b**).* Yellow solid, mp = 156.4−157.4 °C, 42.9 mg (51%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.63 (s, 1H), 8.47 (d, J = 8.0 Hz, 1H), 8.31 (d, J = 8.0 Hz, 1H), 8.16−8.07 (m, 2H), 7.37−7.30 (m, 1H), 7.28 (s, 1H), 7.27−7.19 (m, 3H), 7.12 (dd, J = 15.6, 8.0 Hz, 3H), 6.36 (s, 1H), 3.93 (s, 3H), 1.80 (s, 3H), 0.73 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 183.5, 166.5, 153.6, 147.2, 143.1, 142.0, 133.4, 133.1, 131.4, 131.0, 130.2, 129.3, 128.2, 125.5, 124.6, 124.3, 124.1, 123.2, 122.4, 121.4, 112.2, 110.9, 72.3, 52.6, 27.8, 17.8. HRMS (ESI) m/z calcd for $C_{28}H_{23}NO_3Na$ $[M + Na]^+$: 444.1576, found: 444.1573.

*8-Methoxy-6-(2-methylprop-1-en-1-yl)-6-phenyl-6H-isoindolo-[2,1-a]indole-11-carbaldehyde (**5c**).* Yellow solid, mp = 153.8−154.8 °C, 40.1 mg (51%). Purification (PE/EA = 30/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.56 (s, 1H), 8.27 (d, J = 8.0 Hz, 2H), 7.29−7.27 (m, 2H), 7.26−7.22 (m, 2H), 7.17−7.13 (m, 1H), 7.11−7.08 (m, 2H), 7.07−7.05 (m, 1H), 6.95 (dd, J = 6.4, 2.4 Hz, 2H), 6.40−6.26 (m, 1H), 3.84 (s, 3H), 1.80 (d, J = 1.6 Hz, 3H), 0.82 (d, J = 1.2 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 183.2, 161.7, 156.0, 142.8, 142.6, 133.0, 130.9, 129.2, 128.0, 125.5, 123.4, 122.8, 122.6, 122.2, 113.9, 111.6, 109.7, 109.1, 72.2, 55.8, 27.8, 17.8. HRMS (ESI) m/z calcd for $C_{27}H_{24}NO_2$ $[M + H]^+$: 394.1807, found: 394.1812.

*8-Methyl-6-(2-methylprop-1-en-1-yl)-6-phenyl-6H-isoindolo[2,1-a]indole-11-carbaldehyde (**5d**).* Yellow solid, mp = 198.4−198.8 °C, 41.5 mg (55%). Purification (PE/EA = 50/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.60 (s, 1H), 8.31 (d, J = 8.0 Hz, 1H), 8.19 (s, 1H), 7.32−7.29 (m, 1H), 7.28−7.27 (m, 1H), 7.26−7.24 (m, 2H), 7.23 (s, 2H), 7.20−7.14 (m, 2H), 7.11 (t, J = 1.6 Hz, 1H), 7.09 (dd, J = 2.8, 1.6 Hz, 1H), 6.41−6.30 (m, 1H), 2.40 (s, 3H), 1.79 (d, J = 1.2 Hz, 3H), 0.84−0.67 (m, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 183.3, 154.0, 142.7, 142.5, 140.9, 132.9, 130.8, 129.5, 129.1, 127.9, 126.7, 125.6, 124.1, 124.0, 123.6, 122.9, 122.8, 111.8, 109.8, 72.1, 27.8, 22.1, 17.7. HRMS (ESI) m/z calcd for $C_{27}H_{24}NO$ $[M + H]^+$: 378.1858, found: 378.1862.

*8-Isopropyl-6-(2-methylprop-1-en-1-yl)-6-phenyl-6H-isoindolo-[2,1-a]indole-11-carbaldehyde (**5e**).* Yellow solid, mp = 64.1−66.0 °C, 42.1 mg (52%). Purification (PE/EA = 100/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.60 (s, 1H), 8.32 (d, J = 8.0 Hz, 1H), 8.21 (d, J = 6.4 Hz, 1H), 7.32−7.27 (m, 4H), 7.26−7.20 (m, 2H), 7.19−7.15 (m,1H), 7.13−7.05 (m, 3H), 6.33−6.29 (m, 1H), 3.00−2.90 (m, 1H), 1.79 (d, J = 0.8 Hz, 3H), 1.25 (d, J = 2.0 Hz, 3H), 1.24 (d, J = 2.0 Hz, 3H), 0.77 (d, J = 0.8 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 183.2, 154.0, 151.9, 142.7, 142.3, 133.0, 130.8, 129.1, 127.9, 127.1, 126.8, 125.6, 124.1, 123.6, 123.1, 122.8, 121.8, 111.8, 109.8, 72.4, 34.6, 27.9, 24.1, 23.9, 17.8. HRMS (ESI) m/z calcd for $C_{29}H_{28}NO$ $[M + H]^+$: 406.2171, found: 406.2173.

*8-(tert-Butyl)-6-(2-methylprop-1-en-1-yl)-6-phenyl-6H-isoindolo-[2,1-a]indole-11-carbaldehyde (**5f**).* Yellow solid, mp = 98.3−99.2 °C, 46.9 mg (56%). Purification (PE/EA = 100/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.61 (s, 1H), 8.32 (d, J = 7.6 Hz, 1H), 8.20 (s, 1H), 7.49−7.42 (m, 2H), 7.31−7.28 (m, 1H), 7.27 (d, J = 1.6 Hz, 1H), 7.26−7.20 (m, 2H), 7.19−7.15 (m, 1H), 7.13−7.06 (m, 3H), 6.42−6.20 (m, 1H), 1.80 (s, 3H), 1.32 (s, 9H), 0.77 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 183.2, 154.2, 153.7, 142.8, 142.3, 133.1, 130.7, 129.1, 127.8, 126.7, 125.9, 125.6, 123.6, 123.1, 122.8, 120.5, 111.8, 109.9, 72.5, 35.4, 31.4, 27.9, 17.9. HRMS (ESI) m/z calcd for $C_{30}H_{30}NO$ $[M + H]^+$: 420.2327, found: 420.2331.

*8-Chloro-6-(2-methylprop-1-en-1-yl)-6-phenyl-6H-isoindolo[2,1-a]indole-11-carbaldehyde (**5g**).* Yellow solid, mp = 196.0−196.4 °C, 23.8 mg (30%). Purification (PE/EA = 50/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.57 (s, 1H), 8.49 (s, 1H), 8.29 (d, J = 5.2 Hz, 1H), 7.47−7.43 (m, 2H), 7.43−7.38 (m, 1H), 7.30−7.27 (m, 2H), 7.26−7.21 (m, 2H), 7.09−7.05 (m, 2H), 6.99−6.95 (m, 1H), 6.32−6.30 (m, 1H), 1.79 (d, J = 1.6 Hz, 3H), 0.75 (d, J = 1.2 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 183.0, 153.7, 143.1, 142.2, 132.3, 131.6, 130.6, 129.3, 129.0, 128.7, 128.1, 126.9, 125.1, 124.5, 124.3, 123.6, 122.6, 116.6, 113.2, 109.6, 72.5, 27.8, 17.7. HRMS (ESI) m/z calcd for $C_{26}H_{21}NOCl$ $[M + H]^+$: 398.1312, found: 398.1317.

*10-Chloro-6-(2-methylprop-1-en-1-yl)-6-phenyl-6H-isoindolo-[2,1-a]indole-11-carbaldehyde (**5h**).* Yellow solid, mp = 186.3−186.7 °C, 23.8 mg (30%). Purification (PE/EA = 10/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 11.31 (s, 1H), 8.70 (d, J = 8.0 Hz, 1H), 7.43 (dd, J = 8.0, 1.2 Hz, 1H), 7.37 (dd, J = 7.6, 1.2 Hz, 1H), 7.35−7.30 (m, 3H), 7.27 (d, J = 6.0 Hz, 1H), 7.25−7.17 (m, 4H), 7.07−7.02 (m, 4H), 6.37 (s, 1H), 1.80 (s, 2H), 0.77 (s, 2H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 188.8, 156.1, 145.8, 143.2, 142.5, 132.7, 130.6, 130.4, 130.2, 129.3, 128.9, 128.2, 125.5, 124.6, 124.3, 123.5, 122.9, 122.4, 111.5, 111.1, 71.0, 27.8, 17.8. HRMS (ESI) m/z calcd for $C_{26}H_{21}NOCl$ $[M + H]^+$:398.1312, found: 398.1313.

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*2-Chloro-6-(2-methylprop-1-en-1-yl)-6-phenyl-6H-isoindolo[2,1-a]indole-11-carbaldehyde (**5i**).* Yellow solid, mp = 214.5−214.8 °C, 33.4 mg (42%). Purification (PE/EA = 10/1). ^{1}H NMR (600 MHz, $CDCl_3$) δ 10.58 (s, 1H), 8.43−8.21 (m, 2H), 7.47−7.39 (m, 3H), 7.30−7.21 (m, 3H), 7.15 (dd, J = 8.8, 2.0 Hz, 1H), 7.09−7.05 (m, 2H), 7.04−7.00 (m, 1H), 6.35−6.29 (m, 1H), 1.80 (d, J = 1.2 Hz, 3H), 0.76 (d, J = 0.8 Hz, 3H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 183.0, 153.7, 150.4, 143.0, 142.3, 131.8, 131.3, 130.6, 129.3, 129.1, 129.0, 128.7, 128.1, 125.4, 124.3, 123.7, 122.6, 121.4, 112.7, 109.8, 72.5, 27.8, 17.7. HRMS (ESI) m/z calcd for $C_{26}H_{20}NOClNa$ [M + Na]$^+$: 420.1131, found: 420.1127. It was crystallized from Et_2O/DCM.

*2-Bromo-6-(2-methylprop-1-en-1-yl)-6-phenyl-6H-isoindolo[2,1-a]indole-11-carbaldehyde (**5j**).* Yellow solid, mp = 210.0−211.0 °C, 35.5 mg (42%). Purification (PE/EA = 50/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.57 (s, 1H), 8.49 (s, 1H), 8.28 (s, 1H), 7.48−7.39 (m, 3H), 7.30−7.20 (m, 4H), 7.09−7.03 (m, 2H), 6.97 (d, J = 8.4 Hz, 1H), 6.31 (s, 1H), 1.79 (s, 3H), 0.79 (s, 3H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 182.9, 153.6, 150.2, 143.0, 142.2, 132.2, 131.6, 130.6, 129.2, 129.0, 128.7, 128.1, 126.8, 125.4, 124.4, 124.3, 123.6, 122.5, 116.6, 113.1, 109.6, 72.4, 27.7, 17.7. HRMS (ESI) m/z calcd for $C_{26}H_{20}NOBrNa$ [M + Na]$^+$: 464.0626, found: 464.0627.

*2-Fluoro-6-(2-methylprop-1-en-1-yl)-6-phenyl-6H-isoindolo[2,1-a]indole-11-carbaldehyde (**5k**).* Yellow solid, mp = 187.0−188.0 °C, 42.7 mg (56%). Purification (PE/DCM = 1/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.57 (s, 1H), 8.27 (s, 1H), 8.01 (d, J = 11.2 Hz, 1H), 7.47−7.40 (m, 3H), 7.31−7.20 (m, 3H), 7.10−7.06 (m, 2H), 7.02 (dd, J = 9.2, 4.8 Hz, 1H), 6.95−6.90 (m,1H), 6.38−6.13 (m, 1H), 1.80 (d, J = 1.2 Hz, 3H), 0.76 (d, J = 1.2 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 183.0, 161.1, 158.7, 153.6, 150.6, 143.0, 142.3, 130.4, 129.4, 129.2, 128.7, 128.1, 125.5, 124.2, 123.6, 122.6, 112.6 (d, J = 9.6 Hz), 112.2 (d, J = 26.2 Hz), 110.3, 72.5, 27.8, 17.7. ^{19}F NMR (565 MHz, $CDCl_3$) δ −119.82(1F). HRMS (ESI) m/z calcd for $C_{26}H_{20}NOFNa$ [M + Na]$^+$: 404.1427, found: 404.1426.

*3-Methyl-6-(2-methylprop-1-en-1-yl)-6-phenyl-6H-isoindolo[2,1-a]indole-11-carbaldehyde (**5l**).* Yellow solid, mp = 178.3−178.6 °C, 43.7 mg (58%). Purification (PE/DCM = 1/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.60 (s, 1H), 8.31−8.13 (m, 2H), 7.48−7.35 (m, 3H), 7.31−7.23 (m, 3H), 7.16−7.07 (m, 3H), 6.88 (s, 1H), 6.34−6.32 (m, 1H), 2.39 (s, 3H), 1.80 (d, J = 1.2 Hz, 3H), 0.77 (d, J = 1.2 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 183.3, 154.0, 142.7, 142.5, 140.9, 132.9, 130.8, 129.5, 129.1, 127.9, 126.7, 125.6, 124.1, 124.0, 123.6, 122.9, 122.8, 121.6, 111.8, 109.8, 72.1, 27.8, 22.1, 17.7. HRMS (ESI) m/z calcd for $C_{27}H_{24}NO$ [M + H]$^+$: 378.1858, found: 378.1862.

*(E)-6-(2-Methylbut-1-en-1-yl)-6-phenyl-6H-isoindolo[2,1-a]-indole-11-carbaldehyde (**5m**) and (Z)-6-(2-Methylbut-1-en-1-yl)-6-phenyl-6H-isoindolo[2,1-a]indole-11-carbaldehyde (**5m′**).* Yellow solid (2:1 mixture of isomer), 25.3 mg (34%). Purification (PE/DCM = 1/2). ^{1}H NMR (400 MHz, $CDCl_3$) 10.64 (s, 1H), 8.33 (d, J = 7.2 Hz, 2H), 7.49−7.35 (m, 3H), 7.35−7.15 (m, 5H), 7.14−7.07 (m, 3H), 6.34 (s, 1H × 2/3), 6.32 (s, 1H × 1/3), 2.14−1.98 (m, 2H × 2/3), 1.79(s, 3H × 1/3), 1.22−1.10 (m, 2H × 1/3), 0.99 (t, J = 7.2 Hz, 3H × 2/3), 0.74 (s, 3H × 2/3), 0.22 (t, J = 7.6 Hz, 3H × 1/3). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 183.3, 154.2, 153.7, 149.5, 149.3, 148.3, 148.1, 142.9, 142.8, 132.9, 132.9, 130.8, 130.2, 130.2, 129.4, 129.3, 129.2, 128.5, 127.9, 127.9, 125.9, 125.5, 125.5, 124.3, 124.2, 123.8, 123.6, 123.5, 122.9, 121.6, 112.2, 112.0, 110.2, 72.3, 34.3, 24.6, 24.3, 16.0, 12.9, 10.8. HRMS (ESI) m/z calcd for $C_{27}H_{23}NONa$ [M + Na]$^+$: 400.1677, found 400.1681.

General Procedure for the Synthesis of 6b−6m. To a 50 mL Schlenk tube with a stir bar were successively added **1** (0.5 mmol), **2c** (0.5 mmol, 80 mg), $[Cp^*RhCl_2]_2$ (6.2 mg, 2 mol %), AgOAc (167 mg, 1 mmol), Na_2CO_3 (106 mg, 1 mmol), and EtOAc (3 mL). The reaction was performed at 100 °C in an oil bath for 6 h under an Ar atmosphere. After the reaction was cooled down to room temperature, the crude reaction mixture was filtered through a Celite pad and washed with 20 mL of DCM. The volatiles were removed under reduced pressure, and the residue was subjected to silica gel column chromatography [eluting with petroleum ether (PE)/ethyl acetate (EA)] to afford the corresponding product (**6b−6m**).

*2-(5-Phenylindolo[2,1-a]isoquinolin-6-yl)propan-2-ol (**6b**).* Yellow solid, 140.6 mg (80%).[6a] Purification (PE/EA = 96/4). ^{1}H NMR (600 MHz, $CDCl_3$) δ 8.50 (d, J = 8.4 Hz, 1H), 8.22−8.19 (m, 1H), 7.80 (dd, J = 7.8, 1.2 Hz, 1H), 7.51−7.47 (m, 3H), 7.45−7.42 (m, 2H), 7.35 (dd, J = 7.8, 1.8 Hz, 2H), 7.33−7.30 (m, 1H), 7.27−7.22 (m, 2H), 6.86 (d, J = 8.4 Hz, 1H), 1.68 (s, 6H). It was crystallized from PE/EA.

*2-(9-Methyl-5-phenylindolo[2,1-a]isoquinolin-6-yl)propan-2-ol (**6c**).* Yellow solid, mp = 189.6−189.8 °C, 129.7 mg (71%). Purification (PE/EA = 96/4). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.36 (d, J = 9.2 Hz, 1H), 8.17 (d, J = 8.0 Hz, 1H), 7.56 (s, 1H), 7.50−7.48 (m, 4H), 7.45−7.40 (m, 2H), 7.36−7.30 (m, 3H), 7.23 (t, J = 7.2 Hz, 1H), 7.08 (dd, J = 9.2, 1.6 Hz, 1H), 6.85 (d, J = 8.0 Hz, 1H), 2.52 (s, 3H), 1.64 (s, 5H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 140.8, 139.3, 137.4, 132.9, 132.0, 130.8, 130.4, 130.2, 128.2, 127.7, 127.2, 126.7, 126.4, 125.6, 122.5, 121.6, 121.2, 119.7, 119.5, 95.2, 74.1, 32.8, 21.1. HRMS (ESI) m/z calcd for $C_{26}H_{24}NO$ [M + H]$^+$: 366.1858, found: 366.1852.

*2-(9-Chloro-5-phenylindolo[2,1-a]isoquinolin-6-yl)propan-2-ol (**6d**).* Yellow solid, mp = 179.9−180.3 °C, 131.2 mg (68%). Purification (PE/EA = 96/4). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.63 (s, 1H), 8.17−8.14 (m, 1H), 7.67 (d, J = 8.4 Hz, 1H), 7.50−7.47 (m, 3H), 7.46−7.40 (m, 1H), 7.36 (s, 1H), 7.34−7.26 (m, 3H), 7.23 (dd, J = 7.2, 1.2 Hz, 1H), 6.85 (d, J = 7.6 Hz, 1H), 1.61 (s, 6H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 140.2, 138.9, 137.9, 134.8, 130.3, 128.2, 128.2, 127.9, 127.5, 127.1, 126.6, 125.5, 122.5, 122.2, 122.0, 120.4, 120.3, 95.6, 73.9, 32.9. HRMS (ESI) m/z calcd for $C_{25}H_{21}NOCl$ [M + H]$^+$: 386.1312, found: 386.1303.

*2-(10-Methoxy-5-phenylindolo[2,1-a]isoquinolin-6-yl)propan-2-ol (**6e**).* Yellow solid, mp = 178.9−180.4 °C, 116.4 mg (61%). Purification (PE/EA = 10/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.40 (d, J = 9.6 Hz, 1H), 8.17 (d, J = 7.2 Hz, 1H), 7.51−7.40 (m, 5H), 7.36−7.30 (m, 3H), 7.23 (t, J = 7.2 Hz, 1H), 7.17 (d, J = 2.8 Hz, 1H), 6.89 (dd, J = 9.6, 2.8 Hz, 1H), 6.85 (d, J = 8.0 Hz, 1H), 3.92 (s, 3H), 1.64 (s, 5H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 154.7, 140.5, 137.9, 132.0, 130.8, 130.3, 129.7, 128.2, 127.7, 127.1, 126.7, 126.3, 125.3, 122.4, 121.3, 121.0, 110.1, 100.0, 95.3, 74.0, 55.4, 32.7. HRMS (ESI) m/z calcd for $C_{26}H_{24}NO_2$ [M + H]$^+$: 382.1807, found: 382.1799.

*2-(10-Fluoro-5-phenylindolo[2,1-a]isoquinolin-6-yl)propan-2-ol (**6f**).* Yellow solid, mp = 217.3-217.9 °C, 101.6 mg (55%). Purification (PE/EA = 10/1). ^{1}H NMR (400 MHz, DMSO-d_6) δ 8.72 (dd, J = 10.0, 5.2 Hz, 1H), 8.36 (d, J = 7.6 Hz, 1H), 7.61 (s, 1H), 7.55−7.48 (m, 2H), 7.48−7.42 (m, 1H), 7.37−7.25 (m, 4H), 7.01 (td, J = 9.6, 2.8 Hz, 1H), 6.75 (d, J = 8.0 Hz, 1H), 5.57 (s, 1H), 1.44 (s, 4H). ^{13}C{^{1}H} NMR (100 MHz, DMSO-d_6) δ 158.8, 156.4, 141.6, 138.8, 138.2, 132.0, 131.5, 130.3−130.0 (m), 128.2, 127.8, 127.5, 125.9, 124.6, 123.3 (d, J = 9.1 Hz), 122.9, 120.4, 107.4 (d, J = 25.1 Hz), 103.0 (d, J = 22.2 Hz), 95.7 (d, J = 4.2 Hz), 72.3, 32.7. ^{19}F NMR (565 MHz, $CDCl_3$) δ −122.66 (1F). HRMS (ESI) m/z calcd for $C_{25}H_{21}NOF$ [M + H]$^+$: 370.1607, found: 370.1605.

*2-(10-Bromo-5-phenylindolo[2,1-a]isoquinolin-6-yl)propan-2-ol (**6g**).* Yellow solid, mp = 198.8−199.3 °C, 154.9 mg (72%). Purification (PE/EA = 10/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.41 (d, J = 9.2 Hz, 1H), 8.17 (d, J = 7.6 Hz, 1H), 7.87 (d, J = 2.0 Hz, 1H), 7.49−7.43 (m, 4H), 7.35−7.24 (m, 5H), 6.87 (d, J = 7.6 Hz, 1H), 1.60 (s, 6H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 140.4, 138.9, 138.2, 133.4, 132.0, 131.5, 130.4, 128.2, 127.9, 127.5, 127.3, 126.5, 125.3, 122.6, 122.3, 122.2, 121.9, 114.9, 95.0, 73.9, 32.8. HRMS (ESI) m/z calcd for $C_{25}H_{21}NOBr$ [M + H]$^+$:430.0807, found: 430.0799.

*2-(10-Nitro-5-phenylindolo[2,1-a]isoquinolin-6-yl)propan-2-ol (**6h**).* Red solid, mp = 218.0−218.9 °C, 97.1 mg (49%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, DMSO-d_6) δ 8.79 (d, J = 10.0 Hz, 1H), 8.68 (d, J = 2.8 Hz, 1H), 8.39 (d, J = 7.6 Hz, 1H), 7.97 (dd, J = 10.0, 2.8 Hz, 1H), 7.88 (s, 1H), 7.56−7.50 (m, 4H), 7.42−7.32 (m, 3H), 6.78 (d, J = 8.0 Hz, 1H), 5.63 (s, 1H), 1.44 (s, 6H). ^{13}C{^{1}H} NMR (100 MHz, DMSO-d_6) δ 141.6, 141.4, 139.6, 138.2, 137.4, 131.9, 130.1, 128.7, 128.2, 128.1, 128.1, 126.2, 124.5, 123.2,

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122.4, 122.3, 115.8, 113.1, 97.6, 72.4, 32.8. HRMS (ESI) m/z calcd for $C_{25}H_{21}N_2O_3$ $[M + H]^+$: 397.1552, found: 397.1546.

*6-(2-Hydroxypropan-2-yl)-5-phenylindolo[2,1-a]isoquinoline-10-carbonitrile (**6i**).* Yellow solid, mp = 218.8−219.1 °C, 101.6 mg (54%). Purification (PE/EA = 5/1). ^{1}H NMR (400 MHz, DMSO-d_6) δ 8.80 (d, J = 9.22 Hz, 1H), 8.38 (d, J = 7.2 Hz, 1H), 8.28 (d, J = 2.0 Hz, 1H), 7.73 (s, 1H), 7.55−7.44 (m, 5H), 7.39−7.27 (m, 3H), 6.77 (d, J = 8.0 Hz, 1H), 5.61 (s, 1H), 1.43 (s, 6H). ^{13}C{^{1}H} NMR (100 MHz, DMSO-d_6) δ 141.6, 138.6, 138.3, 136.2, 131.9, 130.2, 129.0, 128.2, 128.1, 127.9, 126.1, 124.9, 124.6, 123.2, 122.8, 122.0, 120.8, 120.3, 103.2, 96.2, 72.4, 32.7. HRMS (ESI) m/z calcd for $C_{26}H_{21}N_2O$ $[M + H]^+$: 377.1654, found: 377.1657.

*Methyl 6-(2-Hydroxypropan-2-yl)-5-phenylindolo[2,1-a]-isoquinoline-3-carboxylate (**6j**).* Yellow solid, mp = 136.8−137.2 °C, 144.8 mg (71%). Purification (PE/EA = 10/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.51 (d, J = 8.8 Hz, 1H), 8.19 (d, J = 8.0 Hz, 1H), 8.02 (dd, J = 8.0, 1.2 Hz, 1H), 7.83−7.79 (m, 1H), 7.57 (d, J = 1.6 Hz, 1H), 7.51 (dd, J = 5.2, 1.6 Hz, 4H), 7.35−7.25 (m, 5H), 3.82 (s, 2H), 1.64 (s, 6H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 166.9, 141.7, 138.4, 136.3, 135.1, 131.9, 130.1, 129.6, 129.1, 128.4, 128.2, 128.1, 127.9, 127.7, 122.4, 121.6, 121.6, 120.6, 120.4, 120.3, 97.8, 74.0, 52.1, 32.8. HRMS (ESI) m/z calcd for $C_{27}H_{24}NO_3$ $[M + H]^+$: 410.1756, found: 410.1748.

*2-(3-Isopropyl-5-phenylindolo[2,1-a]isoquinolin-6-yl)propan-2-ol (**6k**).* Yellow solid, mp = 189.0−189.5 °C, 155.3 mg (79%). Purification (PE/EA = 10/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.49 (d, J = 8.8 Hz, 1H), 8.14 (d, J = 8.4 Hz, 1H), 7.79 (dd, J = 8.0, 1.6 Hz, 1H), 7.51−7.47 (m, 3H), 7.37−7.29 (m, 5H), 7.27−7.21 (m, 1H), 6.70 (d, J = 1.6 Hz, 1H), 2.87−2.77 (m,1H), 1.66 (s, 6H), 1.17 (s, 3H), 1.15 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 147.7, 140.8, 139.3, 137.6, 134.5, 132.0, 130.5, 130.0, 128.1, 127.7, 126.0, 124.0, 123.7, 122.6, 122.0, 121.3, 119.9, 119.9, 119.4, 95.0, 74.1, 34.1, 32.9, 23.8. HRMS (ESI) m/z calcd for $C_{28}H_{28}NO$ $[M + H]^+$: 394.2171, found: 394.2165.

*2-(3-(tert-Butyl)-5-phenylindolo[2,1-a]isoquinolin-6-yl)propan-2-ol (**6l**).* Yellow solid, mp = 198.4−199.1 °C, 166.7 mg (82%). Purification (PE/EA = 10/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.50 (d, J = 8.8 Hz, 1H), 8.14 (d, J = 8.4 Hz, 1H), 7.79 (dd, J = 8.0, 1.2 Hz, 1H), 7.55−7.46 (m, 4H), 7.39−7.34 (m, 3H), 7.33−7.29 (m, 1H), 7.26−7.22 (m, 1H), 6.87 (d, J = 2.0 Hz, 1H), 1.67 (s, 6H), 1.21 (s, 9H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 149.8, 140.7, 139.4, 137.5, 134.5, 132.0, 130.2, 130.0, 128.1, 127.7, 125.1, 123.3, 122.9, 122.3, 121.3, 119.9, 119.5, 95.0, 74.1, 34.8, 32.9, 31.1. HRMS (ESI) m/z calcd for $C_{29}H_{30}NO$ $[M + H]^+$: 408.2327, found: 408.2318.

*2-(5-Phenylindolo[2,1-a]isoquinolin-6-yl)butan-2-ol (**6m**).* Yellow solid, mp = 178.0−178.3. °C, 108.8 mg (60%). Purification (PE/EA = 8/1). ^{1}H NMR (400 MHz, DMSO-d_6) δ 8.55 (d, J = 8.8 Hz, 1H), 8.34 (d, J = 7.6 Hz, 1H), 7.73 (d, J = 7.6 Hz, 1H), 7.59 (s, 1H), 7.53−7.44 (m, 4H), 7.38 (d, J = 6.8 Hz, 1H), 7.32 (d, J = 6.4 Hz, 1H), 7.25 (m, 2H), 7.19−7.13 (m, 1H), 6.72 (d, J = 8.0 Hz, 1H), 5.25 (s, 1H), 2.24 (m, 1H), 1.76 (m, 1H), 1.25 (s, 3H), 0.89 (t, J = 7.2 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, DMSO-d_6) δ 142.2, 139.0, 136.6, 135.3, 132.6, 131.4, 130.1, 129.4, 128.2, 127.9, 127.7, 127.3, 127.0, 125.8, 125.1, 122.6, 121.4, 121.0, 120.8, 119.4, 119.1, 95.6, 74.7, 33.3, 31.1, 7.8. HRMS (ESI) m/z calcd for $C_{26}H_{24}NO$ $[M + H]^+$: 366.1858, found 366.1850.

4 mmol-Scale Experiment of 6j. A mixture of **1j** (4 mmol, 1.015 g), **2c** (4 mmol, 641 mg), $[Cp^*RhCl_2]_2$ (2 mol %, 49.4 mg), AgOAc (8 mmol, 2 equiv, 1.335 g), and EtOAc (23.5 mL) in a 100 mL Schlenk tube was heated in an oil bath at 100 °C for 24 h under an Ar atmosphere. The reaction mixture was cooled to room temperature and concentrated under reduced pressure, and then the residue was subjected to silica gel column chromatography [eluting with petroleum ether (PE)/ethyl acetate (EA)] to afford the corresponding product **6j** (835 mg, 51% yield).

General Procedure for the Synthesis of 7b−7m. A mixture of **6** (0.2 mmol), $TsOH·H_2O$ (0.8 mmol, 4 equiv, 152 mg), $KHSO_4$ (0.8 mmol, 4 equiv,109 mg), and HOAc (1 mL) in a 20 mL Schlenk tube was stirred at 70 °C in an oil bath for 8 h under an air atmosphere. The reaction was quenched with saturated Na_2CO_3 solution (15 mL). The aqueous phase was extracted with ethyl acetate (3 × 20 mL), then the organic phases were combined and washed with water (30 mL), dried over through anhydrous Mg_2SO_4, and concentrated under reduced pressure. The residue was subjected to silica gel column chromatography [eluting with petroleum ether (PE)/ethyl acetate (EA)] to afford the corresponding product (**7b−7m**).

*5-Phenyl-6-(prop-1-en-2-yl) indolo[2,1-a]isoquinoline (**7b**).* Yellow solid, mp = 293.4−294.5 °C, 55.6 mg (83%). Purification (PE/EA = 150/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.26 (d, J = 8.0 Hz, 1H), 8.19 (d, J = 8.8 Hz, 1H), 7.85 (d, J = 8.0 Hz, 1H), 7.49−7.26 (m, 9H), 7.25−7.20 (m, 1H), 7.02 (d, J = 8.0 Hz, 1H), 5.41 (s, 1H), 5.23 (s, 1H), 2.05 (s, 3H).^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 139.3, 137.4, 135.9, 132.9, 132.3, 130.4, 129.8, 128.3, 128.1, 127.4, 127.3, 127.0, 126.5, 125.1, 123.3, 122.5, 122.0, 120.7, 120.5, 119.8, 115.1, 94.0, 23.7. HRMS (ESI) m/z calcd for $C_{25}H_{20}N$ $[M + H]^+$: 334.1596, found: 334.1589.

*9-Methyl-5-phenyl-6-(prop-1-en-2-yl)indolo[2,1-a]isoquinoline (**7c**).* Yellow solid, mp = 289.2−291.4 °C, 45.3 mg (65%). Purification (PE/EA = 150/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.23 (d, J = 8.0 Hz, 1H), 8.06 (d, J = 8.8 Hz, 1H), 7.62 (s, 1H), 7.48−7.40 (m, 4H), 7.36 (d, J = 6.8 Hz, 1H), 7.29 (d, J = 9.2 Hz, 3H), 7.08−6.99 (m, 2H), 5.40 (s, 1H), 5.21 (s, 1H), 2.53 (s, 3H), 2.04 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 139.3, 137.5, 135.9, 132.9, 131.4, 130.6, 130.5, 130.4, 128.3, 128.0, 127.2, 126.9, 126.4, 125.2, 123.2, 122.4, 122.3, 120.0, 119.4, 114.7, 93.6, 23.6, 21.6. HRMS (ESI) m/z calcd for $C_{26}H_{22}N$ $[M + H]^+$: 348.1752, found: 348.1746.

*9-Chloro-5-phenyl-6-(prop-1-en-2-yl)indolo[2,1-a]isoquinoline (**7d**).* Yellow solid, mp = 277.0−278.4 °C, 65.8 mg (89%). Purification (PE/EA = 150/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.25−8.16 (m, 2H), 7.74 (d, J = 8.8 Hz, 1H), 7.46 (dd, J = 18.0, 7.2 Hz, 4H), 7.38−7.21 (m, 5H), 7.02 (d, J = 8.0 Hz, 1H), 5.45 (s, 1H), 5.25 (s, 1H), 2.04 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 138.8, 137.1, 137.0, 136.5, 132.7, 132.3, 130.4, 130.3, 128.3, 128.2, 128.1, 127.7, 127.4, 127.2, 126.6, 125.0, 123.3, 123.0, 122.6, 121.1, 120.4, 114.8, 94.1, 23.5. HRMS (ESI) m/z calcd for $C_{25}H_{19}NCl$ $[M + H]^+$: 368.1206, found: 368.1205.

*10-Methoxy-5-phenyl-6-(prop-1-en-2-yl)indolo[2,1-a]-isoquinoline (**7e**).* Yellow solid, mp = 244.5−245.3 °C, 44.2 mg (61%). Purification (PE/EA = 100/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.23 (d, J = 7.6 Hz, 1H), 8.07 (d, J = 9.2 Hz, 1H), 7.48−7.41 (m, 4H), 7.35 (d, J = 6.8 Hz, 1H), 7.33−7.22 (m, 4H), 7.03 (d, J = 8.0 Hz, 1H), 6.87 (dd, J = 9.6, 2.4 Hz, 1H), 5.40 (s, 1H), 5.22 (s, 1H), 3.93 (s, 3H), 2.03 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 155.4, 139.2, 137.5, 137.2, 136.3, 132.8, 130.6, 130.4, 130.3, 128.2, 128.0, 127.4, 127.2, 126.8, 126.4, 124.8, 123.2, 122.4, 119.3, 115.9, 111.2, 101.0, 93.8, 55.7, 23.5. HRMS (ESI) m/z calcd for $C_{26}H_{22}NO$ $[M + H]^+$: 364.1701, found: 364.1695.

*10-Fluoro-5-phenyl-6-(prop-1-en-2-yl)indolo[2,1-a]isoquinoline (**7f**).* Yellow solid, mp = 294.2−295.5 °C, 49.0 mg (71%). Purification (PE/EA = 200/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.24 (d, J = 8.0 Hz, 1H), 8.12 (dd, J = 9.2, 4.8 Hz, 1H), 7.50−7.41 (m, 5H), 7.37−7.30 (m, 3H), 7.25 (d, J = 1.6 Hz, 1H), 7.04 (d, J = 8.0 Hz, 1H), 6.98−6.92 (m, 1H), 5.42 (s, 1H), 5.23 (s, 1H), 2.03 (s, 3H).^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 139.2, 137.4 (d, J = 3.4 Hz), 137.3, 132.9, 130.6 (m), 129.1, 128.3, 128.1, 127.7, 127.4, 127.1, 126.6, 124.9, 123.4, 122.7, 120.1, 116.0 (d, J = 9.5 Hz), 109.1 (d, J = 26.1 Hz), 104.7 (d, J = 23.0 Hz), 94.1 (d, J = 4.6 Hz), 23.5. ^{19}F NMR (565 MHz, $CDCl_3$) δ −121.49 (d, J = 8.8 Hz, 1F). HRMS (ESI) m/z calcd for $C_{25}H_{19}NF$ $[M + H]^+$: 352.1502, found: 352.1495.

*10-Bromo-5-phenyl-6-(prop-1-en-2-yl)indolo[2,1-a]isoquinoline (**7g**).* Yellow solid, mp = 272.8−273.8 °C, 51.0 mg (62%). Purification (PE/EA = 200/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.23 (d, J = 8.0 Hz, 1H), 8.05 (d, J = 9.2 Hz, 1H), 7.95 (d, J = 1.6 Hz, 1H), 7.50−7.42 (m, 4H), 7.40−7.20 (m, 5H), 7.04 (d, J = 8.0 Hz, 1H), 5.41 (s, 1H), 5.23 (s, 1H), 2.01 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 138.9, 137.1, 137.1, 136.8, 132.7, 131.4, 130.5, 130.3, 128.1, 127.9, 127.4, 127.2, 126.6, 124.8, 123.4, 123.4, 122.8, 122.7, 120.4, 116.4, 115.5, 93.5, 23.5. HRMS (ESI) m/z calcd for $C_{25}H_{19}NBr$ $[M + H]^+$: 412.0701, found: 412.0705.

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*10-Nitro-5-phenyl-6-(prop-1-en-2-yl)indolo[2,1-a]isoquinoline (**7h**).* Red solid, mp = 266.5−268.0 °C, 56.0 mg (74%). Purification (PE/DCM = 5/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.76 (d, J = 2.4 Hz, 1H), 8.26 (dd, J = 7.2, 4.8 Hz, 2H), 8.07 (dd, J = 9.6, 2.4 Hz, 1H), 7.55−7.43 (m, 5H), 7.40−7.33 (m, 2H), 7.25−7.23 (m, 1H), 7.06 (d, J = 8.0 Hz, 1H), 5.47 (s, 1H), 5.28 (s, 1H), 2.03 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 142.9, 138.7, 138.5, 136.7, 136.6, 134.5, 132.5, 130.4, 130.0, 128.9, 128.5, 128.5, 127.7, 127.7, 126.9, 124.5, 123.7, 123.4, 122.0, 117.1, 115.4, 115.0, 95.9, 23.5. HRMS (ESI) m/z calcd for $C_{25}H_{19}N_2O_2$ [M + H]$^+$: 379.1447, found: 379.1449.

*5-Phenyl-6-(prop-1-en-2-yl)indolo[2,1-a]isoquinoline-10-carbonitrile (**7i**).* Yellow solid, mp = 286.1−287.0 °C, 35.5 mg (50%). Purification (PE/EA = 100/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.26 (d, J = 9.2 Hz, 2H), 8.20−8.15 (m, 1H), 7.54−7.33 (m, 8H), 7.25−7.23 (m, 1H), 7.06 (d, J = 8.0 Hz, 1H), 5.45 (s, 1H), 5.25 (s, 1H), 2.02 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 138.7, 137.8, 136.8, 136.7, 133.4, 132.5, 130.5, 130.1, 129.2, 128.4, 128.4, 128.2, 127.6, 126.8, 125.8, 124.6, 123.6, 123.2, 122.8, 121.6, 120.6, 115.7, 104.9, 94.5, 23.5. HRMS (ESI) m/z calcd for $C_{26}H_{19}N_2$ [M + H]$^+$: 359.1548, found: 359.1540.

*Methyl 5-Phenyl-6-(prop-1-en-2-yl)indolo[2,1-a]isoquinoline-3-carboxylate (**7j**).* Yellow solid, mp = 246.1−246.9 °C, 47.7 mg (61%). Purification (PE/EA = 50/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.27 (d, J = 8.3 Hz, 1H), 8.21 (d, J = 8.6 Hz, 1H), 8.09 (dd, J = 8.3, 1.4 Hz, 1H), 7.87 (d, J = 7.9 Hz, 1H), 7.75−7.71 (m, 1H), 7.46 (dd, J = 10.1, 5.4 Hz, 4H), 7.36 (t, J = 7.1 Hz, 2H), 7.30−7.24 (m, 2H), 5.43 (s, 1H), 5.24 (s, 1H), 3.85 (s, 3H), 2.05 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 167.0, 139.0, 138.1, 136.6, 132.8, 132.5, 130.3, 130.1, 129.6, 128.6, 128.5, 128.4, 128.2, 128.1, 127.6, 127.5, 123.3, 122.7, 122.2, 121.5, 119.6, 115.1, 96.2, 52.3, 23.6. HRMS (ESI) m/z calcd for $C_{27}H_{22}NO_2$ [M + H]$^+$: 392.1651, found: 392.1646.

*3-Isopropyl-5-phenyl-6-(prop-1-en-2-yl)indolo[2,1-a]isoquinoline (**7k**).* Yellow solid, mp = 225.2−226.0 °C, 62.4 mg (83%). Purification (PE/EA = 200/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.21 (d, J = 8.4 Hz, 2H), 7.85 (d, J = 8.0 Hz, 1H), 7.50−7.43 (m, 3H), 7.42−7.28 (m, 5H), 7.23 (t, J = 7.6 Hz, 1H), 6.88 (s, 1H), 5.42 (s, 1H), 5.23 (s, 1H), 2.92−2.87 (m, 1H), 2.06 (s, 3H), 1.20 (d, J = 6.8 Hz, 6H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 148.4, 139.4, 137.5, 137.4, 136.0, 132.9, 132.2, 130.5, 130.4, 129.9, 128.2, 128.0, 127.2, 125.5, 124.1, 123.5, 123.2, 122.4, 121.8, 120.3, 119.8, 115.0, 93.4, 77.5, 34.4, 24.1, 23.7. HRMS (ESI) m/z calcd for $C_{28}H_{26}N$ [M + H]$^+$: 376.2065, found: 376.2061.

*3-(tert-Butyl)-5-phenyl-6-(prop-1-en-2-yl)indolo[2,1-a]isoquinoline (**7l**).* Yellow solid, mp = 255.0−255.6 °C, 62.3 mg (80%). Purification (PE/EA = 200/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.20 (d, J = 8.4 Hz, 2H), 7.85 (d, J = 7.9 Hz, 1H), 7.55 (dd, J = 8.3, 1.5 Hz, 1H), 7.49−7.27 (m, 7H), 7.22 (t, J = 7.5 Hz, 1H), 7.03 (s, 1H), 5.42 (s, 1H), 5.24 (s, 1H), 2.06 (s, 3H), 1.23 (s, 9H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 150.5, 139.4, 137.5, 137.3, 136.0, 132.8, 132.2, 130.4, 130.2, 129.9, 128.2, 127.9, 127.2, 124.8, 123.1, 122.7, 122.3, 121.8, 120.3, 120.1, 115.0, 93.4, 35.0, 31.3, 23.7. HRMS (ESI) m/z calcd for $C_{26}H_{22}N$ [M + H]$^+$: 390.2222, found: 390.2221.

*E-6-(But-1-en-2-yl)-5-phenylindolo[2,1-a]isoquinoline (**7m**) and (Z)-6-(But-2-en-2-yl)-5-phenylindolo[2,1-a]isoquinoline (**7m′**).* Yellow solid, 64.1 mg (92%). Purification (PE/EA = 200/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.29−8.23 (**7m** + **7m′**, m, 1H × 0.45 + 1H), 8.10 (**7m**, d, J = 8.4 Hz, 1H × 0.45), 7.94 (**7m′**, d, J = 8.8 Hz, 1H), 7.86 (**7m** + **7m′**, d, J = 8.0 Hz, 1H × 0.45 + 1H), 7.49−7.20 (**7m** + **7m′**, m, 10H × 0.45 + 10H), 7.08 (**7m′**, d, J = 7.6 Hz, 1H), 7.04 (**7m**, d, J = 8.0 Hz, 1H × 0.45), 5.76−5.67 (**7m′**, m, 1H), 5.44 (**7m**, s, 1H × 0.45), 5.35 (**7m**, s, 1H × 0.45), 2.57−2.44 (**7m**, m, 1H × 0.45), 2.25−2.12 (**7m**, m, 1H × 0.45), 1.88 (**7m′**, s, 3H), 1.64 (**7m′**, d, J = 6.8 Hz, 3H), 1.07 (**7m**, t, J = 7.6 Hz, 3H × 0.45). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 144.6, 139.2, 137.8, 137.4, 135.9, 133.1, 132.6, 132.3, 131.2, 130.7, 130.6, 130.5, 130.3, 129.8, 128.1, 128.0, 127.9, 127.4, 127.3, 127.2, 127.0, 126.9, 126.7, 126.5, 126.3, 125.1, 123.3, 123.3, 121.9, 121. 8, 120.6, 120.5, 120.4, 120.0, 119.3, 114.9, 114.9, 94.1, 29.9, 28.6, 17.3, 13.6, 10.9. HRMS (ESI) m/z calcd for $C_{26}H_{22}N$ [M + H]$^+$: 348.1752, found 348.1743.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its online Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.4c00102.

Single-crystal X-ray structure determination (S2 to S7); ^{1}H NMR, ^{13}C NMR spectra of all compounds and HRMS spectra of new compounds (S8 to S260); and single-crystal X-ray structure Crystal data for **3a**, **4a**, **5i**, **6b** (PDF)

Accession Codes

CCDC 2308890, 2308923−2308924, and 2309309 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, by emailing data_request@ccdc.cam.ac.uk, or by contacting the Cambridge Crystallographic Data Center, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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